
5G Regulatory and Policy Framework in Emerging Economies

Item Type Conference Paper

Author L. Banda

Author F. Mekuria

Author M. Mzyece

Abstract The rollout of 5G networks presents a major opportunity for digital transformation in emerging economies, yet regulatory and policy challenges threaten to delay progress. This paper analyzes the current 5G regulatory landscape in emerging economies, focusing on spectrum allocation, licensing, institutional capacity, and regional policy coordination. Using qualitative policy analysis and case studies from South Africa, Nigeria, and Kenya, the study highlights disparities in readiness and identifies critical gaps. Key findings underscore the need for cohesive strategies to address infrastructure, governance, and inclusion. The paper offers policy recommendations, including spectrum reform, regional harmonization, and inclusive regulation, to support an enabling environment for 5G development in emerging economies.

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Tags:

Regulation, Technological innovation, Policy, 5G, Framework, Digital transformation, 5G mobile communication, Resource management, Regulators, Digital twins, Emerging Economies, Employment, Focusing, Green products

5G Technology and Impact on Modern Engineering

Item Type Conference Paper

Author P. Kate

Author U. Pacharney

Abstract 5G technology represents a progressive leap in Wi-Fi conversation, providing unheard-of pace, connectivity, and capacity. Its deployment has profound implications for modern engineering, impacting industries such as telecommunications, healthcare, transportation, and clever towns. With speeds up to one hundred instances quicker than 4G and latency as little as 1 millisecond, 5G permits real-time facts trade, fostering improvements like independent vehicles, far-flung surgeries, and the Internet of Things (IoT). Currently, Engineers are challenged to design infrastructure that will assist the dense

community of gadgets related to 5G, necessitating advancements in antenna design, power performance, and statistics security. The era also quickens the adoption of part computing, reducing reliance on centralised facts centres and ensuring quicker processing near the date supply. The transformative impact of 5G extends beyond connectivity. It reshapes the layout and capability of present-day engineering structures, emphasising adaptability, resilience, and scalability. As 5G continues to conform, its function in driving innovation and redefining engineering paradigms becomes increasingly glaring.

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Extra Journal Abbreviation: 2025 4th International Conference on Sentiment Analysis and Deep Learning (ICSADL)

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Transportation, Technological innovation, Internet of Things, Industries, Telecommunications, Wireless communication, 5G mobile communication, Medical services, Wireless fidelity, 5G technology, Telecommunication network reliability, Modern Engineering, Wireless Communication

A Bayesian Game Based Vehicle-to-Vehicle Electricity Trading Scheme for Blockchain-Enabled Internet of Vehicles

Item Type Journal Article

Author S. Xia

Author F. Lin

Author Z. Chen

Author C. Tang

Author Y. Ma

Author X. Yu

Abstract With ever increasing people's awareness of low carbon and environmental protection, electric vehicles are gradually gaining wide popularity. However, the driving endurance of the electric vehicle is the biggest shortage that hinders the fully acceptance of this new vehicle technology. To deal with this shortage, this paper proposed a vehicle-to-vehicle (V2V) electricity trading scheme based on Bayesian game pricing in blockchain-enabled Internet of vehicles (BIOV). Specifically, the Bayesian game is adopted for pricing in the distributed BIOV with incomplete information sharing. The optimal pricing under the linear strategic equilibrium has been obtained which maximizes the utilities of both sides of electricity transaction. The transaction volume is determined from the formulated convex problem that maximizes the social welfare. Then, the pricing game is implemented by the dedicated smart contract. Blockchain guarantees its trustworthiness, security, and

reliability. Finally, the experimental results show that referring to the benchmark of static game with complete information, the proposed Bayesian game with incomplete information can achieve approximate satisfaction of users. The degree of approximation can reach to 98% when the pricing ranges of buyers and sellers are close. Moreover, the proposed scheme has great advantages over the static game with complete information in terms of communication overhead and timeliness in the decentralized IoVs.

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Blockchain, Electric vehicles, Power markets, Smart contracts, Games, Bayes methods, Bayesian game, electricity trading, Internet of vehicles, Pricing, vehicle-to-vehicle

A First Step Towards Holistic Trustworthy Platoons

Item Type Conference Paper

Author A. Shoker

Author P. Moertl

Author R. Robles

Abstract Truck platooning is a form of convoy cooperative driving of connected trucks assisted by a lead truck. The aim is to reduce the fuel and driving costs, improve road safety, and reduce CO2 emission. Being semi-autonomous, platoons must be trustworthy in many perspectives. This paper presents a high-level trustworthy requirements analysis on three key perspectives: driver, communication, and security. In addition, we observed that any trustworthy requirement analysis is incomplete if perspectives are addressed independently. Therefore, we propose a simple holistic methodology that addresses the different perspectives as well as their dependencies, and we exemplify the use of the methodology with two use cases presented in the paper. However, we draw attention to the importance of more research to drive a more exhaustive and validated methodology¹.

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Security, cybersecurity, Road safety, V2X, Costs, ISO Standards, Internet of Things, automotive, Fuels, trustworthiness, Lead

A Multi-Stage Event Detection Model for Video Datasets Nature-Inspired and Optimized Feature-Based Learning Model

Item Type Conference Paper**Author** R. Chadha**Author** S. Singla**Author** M. A. Wassay

Abstract Estimating the state of the item being tracked is part of the Object Tracking process. Visual tracking is a subject in computer vision that has potential applications in a wide variety of emerging and established technical fields, such as robotics, video surveillance, human-computer interaction, autonomous cars, and sports analytics, to name a few. Therefore, visual trackers are used in mobile robots, in the monitoring of traffic, in the vision of robots, and in the forecasting of accidents. When trying to draw conclusions about the motion of an item based on a set of photographs that have been collected, one of the most challenging challenges is known as “tracking,” and it is one of the most important aspects of this endeavor. The size of an item, its position, and its articulation are the three tasks that provide the greatest challenge in computer vision owing to non-visibility changes. This is because there are certain instances in which the appearance of a visual object is not apparent. The object tracking algorithm is able to function well because of the utilization of a number of characteristics, including the Grayscale feature, Gradient feature, Texture feature, Colour feature, and Fusion of several features. This review article presents a comprehensive assessment in order to learn more about the most current studies in this field and to make suggestions for the conduct of future research. The vast majority of the papers that are being considered are concerned with the identification of events of various kinds, including those that take place in real-time, sports, traffic, natural catastrophes, and other topics. The two issues that have received the most research attention in the field of computer vision are object popularity and the ability to localize an object inside a movie. The popularity of an item is a topic that may be discussed alongside its representation, detection, and categorization. The popularity of object detection and its use in 3-D devices has been studied in a variety of issues, such as monitoring of the places in which it is stored in order to extract it. Given the prior function and length of the target item, the objective of visible tracking is to adjust an arbitrary object in any environment. The aim may be accomplished in one of two ways: either by monitoring a single item or by tracking many objects at the same time. The EFS-linear MSVM approach is proposed as a means of identifying many events within video sequences in this work. The primary objective of this research is to provide an appropriate approach for the selection of features and a classification strategy for the identification of multiple events in You Tube videos. The frames are improved by histogram equalization, and then GLTP, HOG, and Tamura features are employed

to extract the feature vectors, which are often of a higher dimensional type. An effective method of feature selection known as EFS is recommended for use in the process of picking the most useful feature subsets from among the extracted feature vectors. After that, the MSVM is provided with the best feature sub-sets in order to do multi-class classification (20 action scenarios on each database). In the end, the Euclidean distance is used to obtain the events and activities that are relevant. Item tracking is utilized as a result to identify the object in a video and accurately interpret its motion as a sequence of trajectories.

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Tags:

Feature extraction, Visualization, Computer vision, Target tracking, Real-time systems, Computational modeling, Event Detection, Object Tracking, Video event, Video Event Detection, Video sequences, Video Stream, Visual Tracking

A Novel Approach of Data-Driven Strategic Decision-Making in Management: AI-Enabled Analysis of Market Trends, Competitive Intelligence, and Internal Performance Data

Item Type Conference Paper

Author S. Umamaheswari

Author A. Valarmathi

Author D. Paul Dhinakaran

Author C. Vijai

Author S. Sathyakala

Author M. R. Lakshmi

Abstract The term “artificial intelligence” (AI) refers to a wide range of technologies that provide businesses with a variety of benefits, which in turn contribute to increased income. Artificial intelligence (AI) has become increasingly popular among organizations in recent years as a means of generating economic value. This trend can be attributed to the abundance of data and the significant rise in processing capabilities that have occurred in recent years. However, businesses are still facing challenges in implementing and utilizing AI in their normal life operations. The entire influence of AI is examined in this paper, from invention and research through application. The effect of eminent scholastic accomplishments and improvements in the field of artificial intelligence

(simulated intelligence) on business venture and the overall market is examined in this review. The examination likewise makes a commitment to the investigation of the factors driving simulated intelligence's turn of events. The examination of computer based intelligence related innovative undertakings is directed by two records containing the main 200 man-made intelligence new businesses. The examination's determinations will prompt a superior comprehension of the headways and impacts of artificial intelligence on endeavors and society at large. Furthermore, a more profound understanding of how computer based intelligence can reform organization tasks and the worldwide economy will be acquired.

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Extra Journal Abbreviation: 2024 Ninth International Conference on Science Technology Engineering and Mathematics (ICONSTEM)

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Tags:

Artificial intelligence, Network security, Decision making, Technological innovation, Reviews, Artificial Intelligence (AI), Market research, Management, Medical services, Competitive Intelligence, Data Driven Strategy, Mathematics

A Novel Dual FedAux-based Harvested Energy Prediction for Road Safety Systems

Item Type Conference Paper

Author P. S. M. Traoré

Author W. Jaafar

Author F. Baldé

Abstract In the evolving landscape of road safety, the role of solar-powered Internet-of-Things (IoT) devices, crucial for supporting connected and autonomous vehicles (CAVs) with enhanced computing and storage capabilities, is increasingly significant. These devices, often deployed in remote areas such as transcontinental highways, rely heavily on their energy harvesting capabilities for optimal operation. Given their expected widespread deployment, this paper explores the challenge of accurately predicting harvested energy in such IoT devices. To do so, we introduce "Dual FedAux", a novel approach that blends Federated Learning with a dual-model architecture, integrating long short-term memory (LSTM) and Transformer models. This combination, which is enhanced with meta-learning, i.e., stacking ensemble learning, is particularly adapted for time series analysis, such as for energy prediction. Through experiments, we demonstrate that the proposed Dual FedAux significantly outperforms benchmark models in terms of

root-mean-square error (RMSE) and convergence, thus underscoring its potential to enhance the efficiency and reliability of large-scale road safety IoT-based systems.

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Tags:

Road safety, Computer architecture, IoT, Federated Learning, Energy harvesting, Predictive models, Computational modeling, Transformers, Time series analysis, energy prediction, FedAux, Solar energy

A security-aware safety management framework for IoT-integrated bikes

Item Type Conference Paper

Author M. A. Razzaque

Author S. Clarke

Abstract Bike and vehicle collisions often result in fatality to vulnerable bikers. Use of technologies can protect such vulnerable road users. Next generation smart bikes with sensing, computing and communication capabilities or bikes with bikers' smartphones have the potential to be integrated in an Internet of Things (IoT) environment. Unlike avoidance of inter-vehicle collisions, very limited efforts are made on IoT-integrated bikes and vehicles to avoid bike-vehicle collisions and offer bikers' safety. Moreover, these IoT-integrated bikes and vehicles will create new and different information and cyber security risks that could make existing safety solutions ineffective. To exploit the potential of IoT in an effective way, especially in bikers' safety, this work proposes a security-aware bikers' safety management framework that integrates a misbehavior detection scheme (MDS) and a collision prediction and detection scheme (CPD). The MDS, in particular for vehicles (as vehicles are mainly responsible for most bike-vehicle collisions) provides security-awareness to the framework using in-vehicle security checking and vehicles' mobility-patterns-based misbehavior detection. The MDS also includes in-vehicle driver's behavior monitoring to identify potential misbehaving drivers. The framework's MDS and the CPD relies on the improved versions of some existing solutions. Use cases of the framework demonstrates its potential in providing bikers safety.

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Security, Vehicles, V2X communication, Cloud computing, Roads, Trajectory, Estimation, Bikers' Safety, Bikes, Collision Prediction and Detection

A Socio-Technical Approach for Resilient Connected Transportation Systems in Smart Cities

Item Type Journal Article

Author T. Roy

Author A. Tariq

Author S. Dey

Abstract Connected transportation systems in smart cities can be regarded as Cyber-Physical-Social Systems (CPSSs) or Socio-Technical Systems (STSs) due to the presence of connectivity and complex interaction between human and technological infrastructure. Safety and security are essential requirements for such transportation CPSS. Specifically, resilience against cyber-attacks is crucial to address the vulnerabilities of Information and Communication Technologies (ICT) within the infrastructure. In this work, we explore a cyber-attack detection paradigm that leverages the socio-technical nature of such transportation CPSS. Essentially, we exploit the redundancies between physical signals (received from vehicle and infrastructure sensors) and social signals (received from consumers' mobile devices and social media) to detect cyber-attack occurrences. The proposed scheme is developed combining system and control theoretic tools and natural language processing techniques. A case study of a vehicular platoon is considered on which extensive simulation studies have been performed. Such simulation results under different types of cyber-attacks illustrate the promising nature of the proposed approach.

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A Systematic Review of Human–Computer Interaction and Explainable Artificial Intelligence in Healthcare With Artificial Intelligence Techniques

Item Type Journal Article

Author M. Nazar

Author M. M. Alam

Author E. Yafi

Author M. M. Su'ud

Abstract Artificial intelligence (AI) is one of the emerging technologies. In recent decades, artificial intelligence (AI) has gained widespread acceptance in a variety of fields, including virtual support, healthcare, and security. Human-Computer Interaction (HCI) is a field that has been combining AI and human-computer engagement over the past several years in order to create an interactive intelligent system for user interaction. AI, in conjunction with HCI, is being used in a variety of fields by employing various algorithms and employing HCI to provide transparency to the user, allowing them to trust the machine. The comprehensive examination of both the areas of AI and HCI, as well as their subfields, has been explored in this work. The main goal of this article was to discover a point of intersection between the two fields. The understanding of Explainable Artificial Intelligence (XAI), which is a linking point of HCI and XAI, was gained through a literature review conducted in this research. The literature survey encompassed themes identified in the literature (such as XAI and its areas, major XAI aims, and XAI problems and challenges). The study's other major focus was on the use of AI, HCI, and XAI in healthcare. The poll also addressed the shortcomings in XAI in healthcare, as well as the field's future potential. As a result, the literature indicates that XAI in healthcare is still a novel subject that has to be explored more in the future.

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Security, Artificial intelligence, machine learning, Human computer interaction, human-computer interaction, user-centered design, deep learning, explainable artificial intelligence, Data models, Systematics, Medical services, Usability, healthcare, human-centered design, usability

Accelerating Low-Cost Edge-Based Real-Time Video Analytics using Task Scheduling

Item Type Conference Paper

Author K. S. H. Ong

Author D. Niyato

Author C. So-In

Author T. Friedrichs

Abstract Reliable object detection is essential for smart surveillance systems, and video surveillance technology has been significantly improved thanks to Deep-Learning related techniques. The deployment of resource-constraint edge device algorithms based on Deep-Learning objection detection is an attractive approach with a range of challenging practical issues to solve, such as the enforcement of Deep-Learning model inference reliability on inexpensive, general-purpose edge devices. In this paper, we transform the state-of-the-art convolutional Single Shot Detector Object-Detection algorithm into a task optimization problem to investigate whether the overall system reliability can be further improved by using an under-studied technique called semi-partitioned rate-monotonic scheduling algorithm. Unlike current non-scheduling based methods, our scheduling approach and system configuration is empirically shown to achieve state-of-the-art results, such as enhancing Real-Time video inference performance from 0.5 frame-per-second (fps) to 2fps (i.e. four-fold increase) with a recorded mean 68% improvement in video fps deviation. Our findings are important as fluctuations in video fps, hereby termed system reliability, is known to affect the detection accuracy of object detection algorithms. Finally, we discuss the results and highlight several observations that have not been explicitly addressed in current literature.

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Tags:

Object detection, edge computing, Video surveillance, Performance evaluation, Streaming media, Detectors, fixed-priority scheduling, Image edge detection, object-detection, real-time video analytics, Visual analytics

Adaptive Multiplicity Codes based PIR Protocol for Multi-Cloud Platform Services

Item Type Conference Paper

Author L. Salaun

Author A. Alloum

Author P. Jacquet

Abstract In the current paper we proposed a Multiplicity codes based adaptive PIR protocol that is beneficial for multi cloud platform services featured by a high variability in terms of resources, as servers availabilities and storage capacity. Besides the proposed protocol can handle occasional updates when database size varies, with guaranteed minimal communication overhead compared to the trivial case of PIR.

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Tags:

Security, Cloud computing, Protocols, Servers, Databases, Encoding, Frequency modulation

Adopting WirelessHART for In-vehicle-Networking

Item Type Conference Paper

Author Z. Bi

Author D. Chen

Author C. Wang

Author C. Jiang

Author M. Chen

Abstract It is estimated that the breakthrough in the broad deployment of Internet of Things (IoT) could come from smart cars. Indeed, we have seen multi-facet advances around cars: new material, in vehicle infotainment, driverless cars, smart transportation, electrical vehicles, etc. However, in-vehicle-networking has been mainly by wire, the wiring for a car is largely pre-built during the design phase. With more and more things networked within a car, wiring has taken up 1-2 percent of the total weight. This translates into burning up to 0.1 kilogram fuel over 100 kilometers. On the other hand, the advances in wireless technology, especially the broad acceptance of WirelessHART in the industrial settings, has proved its capability in harsh environments. This paper studies what could happen if we use WirelessHART mesh network for in-vehicle communication. While new wireless network protocols are needed to perform the task of CAN, the dominant in-vehicle fieldbus, WirelessHART could take on the work performed by LIN, the fieldbus for peripheral devices. A detailed study is provided to compare these buses. Road tests were performed, in which a WirelessHART network keeps running for the whole 20 minute period.

Date 24-26 Aug. 2015

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Extra Journal Abbreviation: 2015 IEEE 17th International Conference on High Performance Computing and Communications, 2015 IEEE 7th International Symposium on Cyberspace Safety and Security, and 2015 IEEE 12th

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Vehicles, Protocols, Standards, Wireless communication, Wireless sensor networks, Communication system security, CAN, LIN, Reliable Wireless Sensor Network, Smart Car, WirelessHART, Wires

Advancing Public Safety Through Emerging Technologies: a Holistic Approach to Innovation

Item Type Conference Paper

Author R. D. Drakshpalli

Abstract This paper presents an interdisciplinary exploration of emerging technologies with a focus on their application in enhancing public safety. The research addresses several key technological domains and their integration into public safety systems, with emphasis on both operational effectiveness and personnel well-being. Key areas discussed include Artificial Intelligence (AI) and Machine Learning (ML) for crime prediction, decision-making, and pattern detection. Smart Algorithms that automate operational responses and optimize resource allocation. Digital Twins used for emergency simulation and training environments. Intelligent Systems that coordinate various data inputs for improved situational awareness. Edge and Cloud Computing enabling real-time analytics and distributed data processing. Internet of Things (IoT) for real-time sensor and device connectivity across public safety networks. Blockchain for immutable data integrity, forensic logging, and secure communication. Forensics and Cybersecurity tools to safeguard and audit public safety data infrastructure. and Health and Wellness Technologies to monitor and support public safety personnel. Through a comprehensive literature review and a structured methodology, the paper identifies existing gaps in traditional public safety systems. A highlevel solution the Intelligent Public Safety Ecosystem (IPSE) is proposed, which leverages integrated, intelligent infrastructure across domains. The paper analyses case examples demonstrating how these technologies improve operational efficiency, trust, and responsiveness. It also discusses challenges and future directions to create scalable, human-centered, and ethically responsible public safety systems.

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Artificial intelligence, Cybersecurity, V2X, AI, Blockchain, Personnel, IoT, Systematic literature review, Edge Computing, Federated learning, Internet of Things, blockchain, Federated Learning, Public security, Real-time systems, Ethics, Blockchains, ML, Digital twins, blockchain forensics, blockchain-based forensic framework, Communication Networks, digital forensics, Digital Twins, Public Safety, systematic literature review, V2V, Wellness

AI Assurance for the Public – Trust but Verify, Continuously

Item Type Conference Paper

Author P. Laplante

Author R. Kuhn

Abstract Artificial intelligence (AI) systems are increasingly seen in many public facing applications such as self-driving land vehicles, autonomous aircraft, medical systems and financial systems. AI systems should equal or surpass human performance, but given the consequences of failure or erroneous or unfair decisions in these systems, how do we assure the public that these systems work as intended and will not cause harm? For example, that an autonomous vehicle does not crash or that intelligent credit scoring system is not biased, even after passing substantial acceptance testing prior to release. In this paper we discuss AI trust and assurance and related concepts, that is, assured autonomy, particularly for critical systems. Then we discuss how to establish trust through AI assurance activities throughout the system development lifecycle. Finally, we introduce a “trust but verify continuously” approach to AI assurance, which describes assured autonomy activities in a model based systems development context and includes postdelivery activities for continuous assurance.

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Tags:

Artificial intelligence, artificial intelligence, Computer architecture, explainable AI, Vehicle crash testing, Software, Buildings, Land vehicles, Philosophical considerations, zero trust

AI Forensics

Item Type	Conference Paper
Author	S. Lefcourt
Author	G. Falco
Abstract	Artificial intelligence is now a daily topic of public discussion. Not only are intelligent systems such as autonomous vehicles taking the streets, but recommendation algorithms are shaping human behavior. While AI offers a significant increase in efficiency, it can also indirectly cause harm to humans by replacing jobs, violating privacy and even threatening autonomy. To keep track of these cases in which AI has negative implications on a human, databases of AI incidents have been created. We extend this idea of AI incidents to not only include times whereby an AI system caused a real-world harm, but also when it introduces a benefit. Prior work in adjacent fields has defined taxonomies and standard procedures for root cause analysis, digital forensics, AI risk management, and more. Despite these frameworks, there is no means to investigate an AI system to discover the root cause of an incident. We aim to evaluate the body of knowledge that leads to introducing the field of AI Forensics. AI forensics can serve as a postmortem analysis of AI incidents to discover the primary harm catalyst.
Date	6-8 June 2023
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Tags:

AI, Risk management, Privacy, Artificial Intelligence, Taxonomy, Catalysts, Databases, Digital forensics, Forensics, HCI, Human-Centered AI, Root cause analysis, Trustworthy AI

AI Integrated Recommender System for Improving e-commerce Experience

Item Type	Conference Paper
Author	N. Tomar
Author	K. Sahu
Author	V. Kumar
Author	A. Srivastava

Abstract With a broad range of applications for users, recommender systems (RSs), which are made to provide individualized content, have become a crucial technological answer. Despite the fact that numerous RSs have been created and successfully used in various domains, traditional AI-driven RSs continue to encounter difficulties such as data sparsity, cold start problems, and diversity maintenance. By combining artificial intelligence (AI) and human intelligence (HI), augmented intelligence (AuI) makes up for their shortcomings while maximizing their individual strengths. It has been demonstrated that combining artificial intelligence (AI) and human intelligence (HI) improves each technology's capabilities and produces better results than utilizing each technology alone. Recommender systems, which provide predictions for a variety of goods to help consumers with personalized suggestions, have grown in popularity and usefulness. These systems guarantee that they provide users with useful and efficient help by concentrating on particular item categories and utilizing strategies intended to produce tailored recommendations. Particularly during the COVID-19 lockdown, online material streaming has become the most popular kind of entertainment in recent years. Prominent streaming services use a variety of product recommendation techniques to draw customers in and make content recommendations that suit their tastes, which increases user retention. Several recommendation systems are used by well-known streaming services like Netflix, Amazon Prime, and Hulu to improve user experience. However, serious problems including the first-rater dilemma, data sparsity, and computing complexity frequently cause these systems to struggle with accuracy and consistency. In information science, security issues have emerged as a crucial subject of user discourse. E-commerce is becoming increasingly popular as a fundamental part of the Information Science framework. Ironically, though, people are growing more cautious as security and privacy risks increase. Notably, the banking sector has entered a new era as a result of e-commerce, which has changed its services and operations.

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Tags:

Security, Artificial intelligence, Privacy, User experience, Electronic commerce, Human intelligence, Accuracy, Transforms, artificial recommendation System, Information science, Recommendation engine, Recommender systems, smart recommendation in B2C

AI TECH: To Predict Future Pandemics

Item Type Conference Paper**Author** S. Singh**Author** N. Singh**Author** A. Saxena**Author** G. Singh

Abstract The increasing frequency and severity of pandemics across the globe necessitate advanced tools to predict and mitigate future outbreaks. This review paper explores the role of Artificial Intelligence (AI) in forecasting pandemics by analyzing massive, diverse datasets like health records, environmental changes, and even social media posts. AI technologies can enable early detection of a potential threat, along with identification of hot spots, facilitating prompt proactive measures to contain the threat. Machine learning algorithms are instrumental in modeling the spread of infectious diseases, enabling public health authorities to make timely interventions and strategic resource allocation. The paper also highlights the challenges of using AI, including data accuracy and privacy, as well as the ethical concerns around relying on AI for such critical decision-making. The review concludes with a discussion on the future directions of AI in pandemic preparedness, emphasizing the need for interdisciplinary collaboration to develop robust, scalable, and ethically sound AI systems that can better anticipate and respond to global health threats.

Date 13-17 Dec. 2025**DOI** 10.1109/iSES67504.2025.00025**Extra** Journal Abbreviation: 2025 IEEE International Symposium on Smart Electronic Systems (iSES)**ISBN** 2832-3602**Pages** 78-80**Proceedings Title** 2025 IEEE International Symposium on Smart Electronic Systems (iSES)**Conference Name** 2025 IEEE International Symposium on Smart Electronic Systems (iSES)**Date Added** 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm

Tags:

Autonomous vehicles, Artificial intelligence, Machine learning, Protocols, Predictive models, data analytics, Feeds, HTTP, Learning (artificial intelligence), machine algorithms, pattern identifications, Prediction methods, Wireless Access in Vehicular Environments

AI Trustworthy: Ethical Challenges and Strategies

Item Type Conference Paper**Author** J. Liu**Author** I. Sandjaja**Author** D. C. Wunsch

Abstract This paper explores the pivotal role of trust in the widespread application of Artificial Intelligence (AI) across various domains. We review AI applications in sectors like energy, healthcare, and autonomous vehicles and discuss the crisis of human trust they face. This paper introduces a novel framework that delineates

the relationship between AI transparency and user trust, highlighting specific industry applications. Through a systematic review of recent literature, we first delve into factors such as emotional response, acceptance, transparency, accuracy, and interpretability that shape human trust in AI. We then underscore the necessity of ethical AI practices and highlight the importance of regulatory measures to keep pace with technological advancements. Finally, we propose strategic measures for maintaining trust in AI, focusing on stringent regulation, ethical authorization, and effective human-AI cooperation, thereby contributing to the ongoing discourse on AI trust.

Date 22-24 June 2024

DOI 10.1109/SOLI63266.2024.10956105

Extra Journal Abbreviation: 2024 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI)

ISBN 2768-1890

Pages 1-6

Proceedings Title 2024 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI)

Conference Name 2024 IEEE International Conference on Service Operations and Logistics, and Informatics (SOLI)

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Tags:

Artificial intelligence, artificial intelligence, Regulation, Logistics, Systematic literature review, ethics, decision-making, Shape, Ethics, Medical services, Informatics, Industry applications, regulation and policy, Shape measurement, trustworthiness

Android Automotive for Software-Defined Vehicles: An Assessment of Capabilities, Limitations, and Future Directions

Item Type Conference Paper

Author J. Stümpfle

Author A. K. V. Narla

Author N. Jazdi

Author M. Weyrich

Abstract The automotive industry is rapidly shifting towards software-defined vehicles (SDVs), with automotive operating systems becoming central to system integration. Android Automotive Operating System (AAOS) has gained significant adoption from manufacturers due to its developer ecosystem, Over-the-Air capabilities, and Hardware Abstraction Layer that decouples software from hardware. Despite its growing industry presence, a gap exists in academic research examining AAOS's role in the SDV transformation. This paper provides a technical assessment of AAOS through code analysis, architecture review, and implementation on an intelligent cockpit platform. We examine AAOS's potential beyond infotainment to more integral roles in automotive architecture, identifying both strengths—software flexibility, scalability, and cloud integration—and limitations—realtime performance constraints, Google ecosystem dependency,

and safety-critical function capabilities. Our contributions include a systematic architecture review focused on HAL implementation, identification of technical limitations for broader automotive applications, documentation of implementation challenges through our case study, and extended architecture components addressing identified limitations. This research informs both academic discourse and industry practice regarding AAOS's role in the evolution towards fully software-defined vehicles.

Date 30 Sept.-2 Oct. 2025

DOI 10.1109/IAVVC61942.2025.11219511

Extra Journal Abbreviation: 2025 IEEE International Automated Vehicle Validation Conference (IAVVC)

Pages 1-6

Proceedings Title 2025 IEEE International Automated Vehicle Validation Conference (IAVVC)

Conference Name 2025 IEEE International Automated Vehicle Validation Conference (IAVVC)

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Tags:

Ecosystems, Automotive engineering, Reviews, Wireless networks, Hardware, Survey, Industries, Software, Systematics, Android Automotive OS, Operating systems, Software Defined Vehicles, Technical management, Vehicle Abstraction Layer

Application of Physiological Sensors for Personalization in Semi-Autonomous Driving: A Review

Item Type Journal Article

Author E. J. C. Nacpil

Author Z. Wang

Author K. Nakano

Abstract As automobiles become increasingly autonomous, there is growing interest in personalized automation based on human behavior. However, there is an increasing need for behavioral data from vehicle occupants that could be acquired through physiological sensors. Hence, the motivation of this review is to provide information to develop practical physiological sensors to assess behavior, while maintaining the acceptance, privacy, and safety of vehicle occupants. In order to improve driver and passenger experiences and build trust between autonomous vehicles and humans, the current review considers a number of studies addressing driving tasks based on sensors that monitor vehicle occupant behavior. By including applications to emergency and non-emergency driving scenarios, a survey of literature is conducted on physiological sensing for personalization of semi-autonomous driving. Given the limited availability of sensor technology that is specifically designed to monitor behavior in automobiles, the survey is provided as a resource to further physiological sensor development for automotive applications.

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DOI 10.1109/JSEN.2021.3100038

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Tags:

autonomous vehicles, Safety, Vehicles, Automobiles, Advanced driver assistance systems, human-machine interaction, Sensors, Production, Monitoring, Biomedical monitoring, biosignal sensors, human behavior, personalization, physiological measurement

Applying E-Payment for Improving MSME's Revenue in Jabodetabek: An Extended TAM Analysis

Item Type Conference Paper
Author E. S. Kolly
Author M. I. Wijaya
Author T. Andrian
Abstract The objective of this research is to explore the variables that influence the usage of E-Payment in Jabodetabek region's MSMEs as well as how it may influence the revenue of the business, using the Extended Technology Acceptance Model. The quantitative method is use to collect data and information, where primary data will be obtained through Online Questioners and interviews with MSMEs owners and employees-in-charge, 188 responses were processed using Smart PLS 4. The findings show that applying E-Payment in MSMEs has a positive and significant impact towards increasing their revenue. However, many respondents reported problems regarding a delay for their revenue reimbursement and some connectivity issues. Thus, not only making it hard to manage their cash flow, but also made some MSMEs not interested in continuing to use E-Payment. This study gave insights to EPayment vendors to improve their platforms so more MSMEs will apply this payment method for their business and help the Indonesian government to fulfill their digital transformation for transactions in the country.
Date 7-8 Nov. 2023
DOI 10.1109/ICCED60214.2023.10425591
Extra Journal Abbreviation: 2023 IEEE 9th International Conference on Computing, Engineering and Design (ICCED)
ISBN 2767-7826
Pages 1-6
Proceedings Title 2023 IEEE 9th International Conference on Computing, Engineering and Design (ICCED)
Conference Name 2023 IEEE 9th International Conference on Computing, Engineering and Design (ICCED)
Date Added 20/09/2026, 7:47:03 pm

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Tags:

Security, Technology acceptance model, Urban areas, Interviews, Computational modeling, Government, Business, E-Payment, MSME Revenue, Technology Acceptance Model

Architecting Communication for Public Safety in Mixed Traffic Environments with Autonomous Driving Systems

Item Type Conference Paper

Author A. Iwamura

Author W. Chen

Author H. Nishimura

Abstract The widespread adoption of autonomous driving systems (ADSs) is expected to reduce accidents caused by human error and enhance public safety. However, during the transitional period, ADS-equipped vehicles will coexist with manually driven vehicles and pedestrians, creating a mixed traffic environment. Discrepancies in communication between ADSs and other travelers may inadvertently introduce public safety risks. To address this challenge, the authors have previously proposed a high-level operational concept that promotes ethical communication among road users based on P.F. Strawson's concept of reactive attitudes. Using this concept, an ADS architecture was defined and the communication requirements for improving public safety in mixed-traffic environments were clarified. In this study, the previously proposed operational concept is further developed by systematizing the entire community as an enterprise. Using the unified architecture framework (UAF), the enterprise architecture is defined to continuously improve public safety across an entire community. The focus is particularly on promoting cooperative behavior among road users based on ethical values and forming a network to spread public safety awareness within a community. This study contributes to a sustainable and safe traffic environment through promoting ethical cooperation between ADSs and road users.

Date 23-25 Sept. 2025

DOI 10.1109/WF-PST65083.2025.00018

Extra Journal Abbreviation: 2025 IEEE World Forum on Public Safety Technology (WF-PST)

Pages 74-81

Proceedings Title 2025 IEEE World Forum on Public Safety Technology (WF-PST)

Conference Name 2025 IEEE World Forum on Public Safety Technology (WF-PST)

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Tags:

Autonomous vehicles, Pedestrians, Autonomous driving system, Interviews, Stakeholders, Roads, Public security, public safety, Ethics, Adaptation models, ethical communication, mixed-traffic environment, Service-oriented architecture, System dynamics, Unified architecture framework

Blockchain based Internet of Vehicles (IOV) InformationTransmission Mechanisms

Item Type Conference Paper

Author L. R

Author S. S

Abstract The automotive industry has certainly reshaped society and becoming sophisticated, intelligent, and highly developed industries with discoveries ranges from hybrid, battery-powered, and self-driving modern vehicles to the advent of the Internet of Vehicles (IoV). Because of its complication, it necessitates the use of numerous Industry 4.0 technology such as modern manufacturing operations, robotics, cyber security, augmented reality and so on. The blockchain is a new trend in the automotive industry that can improve data security, anonymity, traceability, integrity, accountability, privacy, robustness, authentication transparency and trustworthiness. Internet of Vehicles (IoV) relates to the transmission of actual data between infrastructures and vehicles via intelligent terminal devices, information platforms, automotive navigation processes, and mobile telecommunication technologies. The article gives a systematic review of Internet of Vehicles (IoV). The analysis is based on evaluation metrics, limitations, technological advancements and techniques over a span of time. This essence has created an opportunity to collect and distribute information regarding vehicles and infrastructures more easily. It moreover enables data collection, processing, and sharing in social media. Most vehicular messages need to be shared in close proximity vehicles whereas IoV helps to improve traffic effectiveness and security. Some messages may need to be distributed more widely in a larger region. Currently, the technique is generating a lot of research interest in other areas, such as the Internet of Vehicles (IoVs). Some centralized issues can be resolved in the blockchain technology using a decentralized and reliable vehicular environment.

Date 13-15 Oct. 2022

DOI 10.1109/ICECAA55415.2022.9936403

Extra Journal Abbreviation: 2022 International Conference on Edge Computing and Applications (ICECAA)

Pages 514-523

Proceedings Title 2022 International Conference on Edge Computing and Applications (ICECAA)

Conference Name 2022 International Conference on Edge Computing and Applications (ICECAA)

Date Added 20/09/2026, 7:47:04 pm

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Tags:

Social networking (online), Navigation, Robustness, Industries, Systematics, Measurement, Service robots

Blockchain-Assisted Trust Management in Vehicular Ad-Hoc Network

Item Type Conference Paper

Author S. Banerjee

Author P. Kuila

Author B. B. Naik

Abstract Vehicle-to-everything (V2X) communication is crucial in Intelligent Transportation Systems (ITSs) for enhancing road safety, traffic efficiency, and convenience. To address the need for trust in V2X communications, a trust management system for Vehicular Ad Hoc Networks (VANETs) using blockchain technology has been proposed. This system calculates dynamic trust scores for vehicles based on direct communication, recommendations from other vehicles, and proximity to roadside units (RSUs). The trust scores are stored in a transparent and tamperproof blockchain, ensuring the immutability and transparency of trust scores. Simulations demonstrate that this blockchain-based trust management system significantly improves the security and reliability of V2X communications. This system is essential for the widespread acceptance of VANETs by fostering a trusted communication environment.

Date 24-28 June 2024

DOI 10.1109/ICCCNT61001.2024.10724619

Extra Journal Abbreviation: 2024 15th International Conference on Computing Communication and Networking Technologies (ICCCNT)

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Pages 1-7

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Tags:

Trust, Security, Road safety, V2X, Reliability, Blockchain, Vehicular ad hoc networks, Intelligent transportation systems, Trust management, Edge computing, and Edge computing, VANET, Vehicle-to-everything, Vehicle dynamics, Blockchains

Christine the Murderer Artificial Intelligence

Item Type Conference Paper

Author G. Kiss

Abstract We read about the results of artificial intelligence research almost every day in magazines, on social media portals and in news reports. The topic of self-driving vehicles is also coming up more and more often in the media, beyond the professional circles, more often in connection with an accident. The successful implementation of self-driving cars is unthinkable without artificial intelligence. In this article, we present techniques to confuse artificial intelligence in self-

driving vehicles, to force it to make a decision that could cause an accident. At the end of the paper, we outline how to create murderer artificial intelligence in self-driving vehicles that could pose a threat not only to the occupants of the vehicle, but also to their environment.

Date 17-20 May 2022

DOI 10.1109/CoDIT55151.2022.9803913

Extra Journal Abbreviation: 2022 8th International Conference on Control, Decision and Information Technologies (CoDIT)

Volume 1

ISBN 2576-3555

Pages 1427-1431

Proceedings Title 2022 8th International Conference on Control, Decision and Information Technologies (CoDIT)

Conference Name 2022 8th International Conference on Control, Decision and Information Technologies (CoDIT)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Artificial intelligence, artificial intelligence, Decision making, Social networking (online), self-driving vehicles, Hazards, Economics, Media, confused, dangerous, Force, murder

CNN Vulnerability On Untargeted Adversarial Attacks

Item Type Conference Paper

Author A. Stankovic

Author M. Marjanovic

Abstract Convolutional Neural Networks (CNNs) are widely used in various AI-driven industries, including healthcare, autonomous systems, and cybersecurity, due to their high accuracy in image classification. However, these models are vulnerable to adversarial attacks, where small, imperceptible perturbations in input data can cause incorrect predictions. This study evaluates the impact of five adversarial attacks: Fast Gradient Sign Method (FGSM), Iterative FGSM (IFGSM), Momentum Iterative FGSM (MIFGSM), Deepfool, and OnePixel Attack on the accuracy of a CNN trained on the CIFAR-10 dataset. Each attack is analyzed for its unique influence on model performance, revealing significant accuracy drops under adversarial conditions. The results are analyzed and illustrate the potential security risks of adversarial attacks, underscoring the need for robust defenses in CNN based systems.

Date 26-27 Nov. 2024

DOI 10.1109/TELFOR63250.2024.10819181

Extra Journal Abbreviation: 2024 32nd Telecommunications Forum (TELFOR)

ISBN 2994-5828

Pages 1-4

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Conference Name 2024 32nd Telecommunications Forum (TELFOR)

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Convolutional neural networks, adversarial attacks, Iterative methods, Industries, Resilience, Telecommunications, Predictive models, Accuracy, Medical services, Training, neural network, Perturbation methods, model accuracy

Co-operative game for certification and continued conformance check of AI enabled CPS*

Item Type Conference Paper**Author** A. Banerjee**Author** A. Maity**Author** I. Lamrani**Author** K. S. Sandeep Gupta

Abstract One of the biggest challenges in widespread acceptance and deployment of cyber physical systems (CPS) such as medical devices or autonomous cars is generating trust in users about their safety, security and effectiveness. As such the certification process is of utmost importance. However, the process of certification for a CPS is potentially an intractable problem and can never be optimally solved to obtain a CPS that is regulatory compliant in an absolute sense. Introduction of data driven learning introduces non-determinism that further exacerbates the problem. Moreover, the certification process involves collaboration between certifier and manufacturer who are potentially restricted by the law of the land to communicate with full transparency. In this paper, we navigate the certification space in the medical device domain and provide a framework through co-operative gaming to certify a CPS under a limited operational environment in an agile iterative process with assurance guarantees.

Date 12-15 May 2024**DOI** 10.1109/ICPS59941.2024.10639951**Extra** Journal Abbreviation: 2024 IEEE 7th International Conference on Industrial Cyber-Physical Systems (ICPS)**ISBN** 2769-3899**Pages** 1-6**Proceedings Title** 2024 IEEE 7th International Conference on Industrial Cyber-Physical Systems (ICPS)**Conference Name** 2024 IEEE 7th International Conference on Industrial Cyber-Physical Systems (ICPS)**Date Added** 20/09/2026, 7:47:04 pm**Modified** 20/09/2026, 7:47:04 pm**Tags:**

Safety, Security, Cyber-physical systems, Navigation, Collaboration, Cyber-Physical Systems, Games, cooperative game theory, Medical devices, Safety Certification

Common Framework for 5G Northbound APIs

Item Type Conference Paper

Author N. D. Tangudu

Author N. Gupta

Author S. P. Shah

Author B. J. Pattan

Author S. Chitturi

Abstract 3rd Generation Partnership Project (3GPP), a global standards forum for cellular communication technologies, exposed the core network capabilities as Northbound APIs for consumption by 3rd parties outside Mobile Network Operator (MNO) domain such as enterprise domain Application Servers (AS) or Application Function (AF). However, the common supporting capabilities like onboarding, discovery, publishing, authentication, logging, auditing etc., applicable for all Northbound APIs, were not specified in 3GPP until Rel-14. In addition, with the advent of 5G networks, it is natural that more vertical industries look to enable their communications on 5G networks, and so the increase in 3rd parties ecosystem consuming the 5G Northbound APIs. In Rel15, 3GPP specified Common API Framework (CAPIF), to avoid duplication in being specified at multiple places and bring in consistency of approach between different API specifications, and to ease the development and deployment of 3rd party applications invoking 3GPP Northbound APIs for Evolved Packet System (EPS) and 5G System (5GS). In this technical paper, the CAPIF standard for 3GPP Northbound APIs including the common supporting capabilities enabled by the framework are explained. The paper also elaborates how CAPIF can be leveraged to quickly develop and deploy different 3GPP Northbound APIs and the vertical industry applications in unified and harmonized manner over the 5G Network.

Date 10-12 Sept. 2020

DOI 10.1109/5GWF49715.2020.9221161

Extra Journal Abbreviation: 2020 IEEE 3rd 5G World Forum (5GWF)

Pages 275-280

Proceedings Title 2020 IEEE 3rd 5G World Forum (5GWF)

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Tags:

Authentication, Ecosystems, Authorization, Industries, 5G mobile communication, 3GPP Northbound interfaces, 5G networks, 5G Northbound APIs, CAPIF, Communications technology, Publishing

Computational Intelligence and Its Transformative Influence

Item Type Conference Paper

Author J. A P

Author A. Shankar

Author A. Narayan

Author M. T. R

Author P. Gaur

Author S. S. Kumar

Abstract Computational intelligence is reshaping industries such as healthcare, finance, transportation, and manufacturing, with the increasing sophistication of artificial intelligence driving substantial market growth. The Comprehensive Brain Function Simulation (CBFS) Model leverages computational intelligence to study the human brain, advancing our understanding of cognition and enabling the simulation of neurological disorders. Neural networks are pivotal in machine learning, powering tasks like image recognition and decision-making. Game theory models offer insights into rational decision-making, applicable in economics, politics, and social sciences. Agent-Based Financial Models simulate financial markets, analyzing dynamics and trading strategies. This study underscores computational intelligence's transformative potential, emphasizing innovation and problem-solving across diverse sectors. The burgeoning computational intelligence market and the use of models like brain function simulation, machine learning, game theory, and financial analysis herald a new era of possibilities in the field.

Date 5-7 April 2024

DOI 10.1109/I2CT61223.2024.10543715

Extra Journal Abbreviation: 2024 IEEE 9th International Conference for Convergence in Technology (I2CT)

Pages 1-7

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Tags:

Transportation, Machine learning, Healthcare, Neural Networks, Machine Learning, Brain modeling, AI Technologies, Industries, Computational modeling, Medical services, Advanced Algorithms, Brain Function Simulation, Computational Intelligence, Game Theory Model, Optimization Processes, Social sciences

Corporate Social Responsibility Challenges and Risks of Industry 4.0 technologies: A review

Item Type Conference Paper

Author D. O. Mora Sanchez

Abstract The fourth industrial revolution arrived with many enabling technologies that would impact important sociological aspects in the industry. Some of the Industry 4.0 technologies are already running in different industrial application, and other are still as a paradigm state. The social, economic, and environmental acceptance of Industry 4.0 technologies is still under discussion, which open new opportunities to execute various analysis about the possible implications of the implementation of such technologies. This article refers to an exploratory analysis

and identification of the different challenges and risks of this new Industry 4.0 paradigm and its related technologies. The technologies under review were Internet of Things, Artificial Intelligence, Cloud Computing, cybersecurity, big data, blockchain, 5G, robotics, adding manufacturing, unmanned systems, autonomous vehicles, virtual reality, and augmented reality. As a result, different social challenges and risks were identified for each technology, starting from vulnerability, implementation cost, until social aspects such as education and unemployment caused by those new technologies. In conclusion, Industry 4.0 arrived with a lot of benefits to the industry business, but companies should not stop thinking about sustainable development.

Date 4-5 June 2019

Extra Journal Abbreviation: Smart SysTech 2019; European Conference on Smart Objects, Systems and Technologies

Pages 1-8

Proceedings Title Smart SysTech 2019; European Conference on Smart Objects, Systems and Technologies

Conference Name Smart SysTech 2019; European Conference on Smart Objects, Systems and Technologies

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Deep Learning Based Hand Wrist Crease Object Detection Models: An In-Depth Analysis

Item Type Journal Article

Author G. Elumalai

Author G. Malathi

Abstract Deep learning has evolved dramatically in recent years, resulting in a wide range of useful applications in various fields, such as computer vision, natural language processing, robotics, and autonomous vehicles. This expansion demonstrates the high transformative potential of deep learning techniques in real-world applications. This study addresses two critical challenges in this context. First, we recognized the unique potential of hand wrist creases as robust biometric identifiers for individuals. Second, we acknowledge the importance of custom deep learning models in enhancing the accuracy of object detection for hand wrist creases. In response, we propose a novel deep learning method for hand wrist crease object detection by leveraging, custom yolov8 models in this research. Our experiments involved comparing the performance of the custom Yolov8 with the Yolov5 and Yolov8 models. Notably, the custom Yolov8x model achieved an impressive object-detection accuracy of 97.1%. This innovative approach has the potential to significantly affect the development of biometric security systems, particularly in scenarios where traditional biometric identifiers are impractical or unreliable. By harnessing the power of deep learning and custom model architectures, this study paves the way for more accurate and reliable biometric identification systems, contributing to the advancement of security measures in a wide range of applications and domains.

Date 2024

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Publication IEEE Access**ISSN** 2169-3536**Journal Abbr** IEEE Access**Date Added** 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Security, Deep learning, Feature extraction, Training, Biometrics (access control), YOLO, object detection, Hand wrist crease, HWBID, HWC-Yolov8, Wrist

Deep learning in the automotive industry: Applications and tools

Item Type Conference Paper**Author** A. Luckow**Author** M. Cook**Author** N. Ashcraft**Author** E. Weill**Author** E. Djerekarov**Author** B. Vorster

Abstract Deep Learning refers to a set of machine learning techniques that utilize neural networks with many hidden layers for tasks, such as image classification, speech recognition, language understanding. Deep learning has been proven to be very effective in these domains and is pervasively used by many Internet services. In this paper, we describe different automotive uses cases for deep learning in particular in the domain of computer vision. We surveys the current state-of-the-art in libraries, tools and infrastructures (e. g. GPUs and clouds) for implementing, training and deploying deep neural networks. We particularly focus on convolutional neural networks and computer vision use cases, such as the visual inspection process in manufacturing plants and the analysis of social media data. To train neural networks, curated and labeled datasets are essential. In particular, both the availability and scope of such datasets is typically very limited. A main contribution of this paper is the creation of an automotive dataset, that allows us to learn and automatically recognize different vehicle properties. We describe an end-to-end deep learning application utilizing a mobile app for data collection and process support, and an Amazon-based cloud backend for storage and training. For training we evaluate the use of cloud and on-premises infrastructures (including multiple GPUs) in conjunction with different neural network architectures and frameworks. We assess both the training times as well as the accuracy of the classifier. Finally, we demonstrate the effectiveness of the trained classifier in a real world setting during manufacturing process.

Date 5-8 Dec. 2016**DOI** 10.1109/BigData.2016.7841045**Extra** Journal Abbreviation: 2016 IEEE International Conference on Big Data (Big Data)**Pages** 3759-3768**Proceedings Title** 2016 IEEE International Conference on Big Data (Big Data)**Conference Name** 2016 IEEE International Conference on Big Data (Big Data)

Date Added 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Machine learning, Automotive engineering, Neural networks, Cloud Computing, Deep Learning, Computational modeling, Training, Libraries, Manufacturing, Automotive, Sparks

Deep Learning: Techniques, Taxonomy, Applications, and Future Directions

Item Type Conference Paper**Author** V. Mohana Sundari**Author** R. A. Mabel Rose**Author** S. M. Sassirekha**Author** M. Nalini**Author** P. Girija**Author** S. S. R

Abstract Deep learning has become one of the key AI techniques of the modern world and has changed many fields by its capability to learn from data, which is often very complex. The presented survey paper aims at describing fundamental techniques and taxonomy of deep learning, various applications, and potential perspectives for further research. The paper starts with presenting the concept of deep learning and its origins as well as emphasizing the important position of DL in the modern development of AI. It then goes further into deep learning workflow and frameworks and presents feedforward neural networks, CNN, RNN, generative adversarial networks, attention mechanisms, and transformers. An overview of deep learning frameworks is discussed, including supervised, unsupervised, reinforcement, transfer, semi-supervised, & self-supervised learning. The survey further continues to provide a detailed description on how DL is being widely applied in CV, NLP, speech recognition, disease diagnosis & stock price forecasting. The issues related to challenges and limitations include data dependency, interpretability, overfitting, high computational burden, and ethical concerns are presented along with ongoing work and directions for future investigations. This includes explainability of AI, incorporation of symbolic AI methods, improving the resilience and security of models, breakthroughs in hardware, as well as expansion into future domains such as quantum computing and robotics. Finally, this survey paper summarizes existing literature and predicts future research direction, emphasizing on how deep learning will revolutionise technology and how it may foster development in various fields.

Date 6-8 Nov. 2024**DOI** 10.1109/ICSTSDG61998.2024.11026496**Extra** Journal Abbreviation: 2024 International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG)**Pages** 1-5**Proceedings Title** 2024 International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG)**Conference Name** 2024 International Conference on Smart Technologies for Sustainable Development Goals (ICSTSDG)

Date Added 20/09/2026, 7:47:04 pm**Modified** 20/09/2026, 7:47:04 pm**Tags:**

Artificial intelligence, AI, Deep learning, CNN, Deep Learning, Robots, Surveys, Machine Learning, Hardware, Quantum computing, Computational modeling, Ethics, Transformers, RNN, Taxonomy

Deep Neural Networks Study for Advanced Code-reuse Attacks Detection

Item Type Conference Paper**Author** J. M. Oliveira**Author** A. Musa**Author** E. Parisi**Author** F. Barchi**Author** A. Acquaviva

Abstract The RISC-V architecture is becoming increasingly common in embedded systems, including industrial controllers and autonomous vehicles, where security is a critical concern. These systems require protection against cyberattacks that can compromise program flow and trigger malicious behavior. Recent approaches to applying Control Flow Integrity (CFI) in RISC-V systems involve either modifying the instruction set architecture or implementing custom hardware IP. Although effective against many Code Reuse Attacks (CRAs), these countermeasures often depend on specialized toolchains, custom binaries, or dedicated CFI hardware. Deep Learning (DL)-based techniques, on the other hand, can learn complex, high-dimensional signatures of code reuse, and they can incorporate features from any instruction stream or microarchitectural trace. In this work, we propose and evaluate several Deep Neural Network (DNN) architectures for detecting CRAs. Specifically, we propose a framework for dataset creation and supervised and semi-supervised training of DNNs. These DNNs integrate state-of-the-art embedding mechanisms along with recurrent, and attention layers, allowing us to evaluate their performance in recognizing valid execution paths within benchmark programs. Execution traces were extracted using QEMU on RISC-V architectures and using the Intel Processor Trace (IPT) on Intel x86_64 architectures. Our results show that a simple RNN based on GRU layers is sufficient to achieve 99% accuracy in Intel x86_64 and 96% accuracy in RISC-V.

Date 24-26 Aug. 2025**DOI** 10.1109/RTSI64020.2025.11212515**Extra** Journal Abbreviation: 2025 IEEE 9th Forum on Research and Technologies for Society and Industry (RTSI)**ISBN** 2687-6817**Pages** 332-337**Proceedings Title** 2025 IEEE 9th Forum on Research and Technologies for Society and Industry (RTSI)**Conference Name** 2025 IEEE 9th Forum on Research and Technologies for Society and Industry (RTSI)**Date Added** 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Cybersecurity, Deep learning, Computer architecture, Deep Learning, Hardware, Real-time systems, Performance evaluation, Accuracy, Training, Artificial neural networks, Code Reuse Attacks Detection, Instruction sets, Instruction Traces, RISC-V, Tokenization

Deepfake Detection using Inception-ResNet-V2 Network

Item Type Conference Paper

Author R. R. Rajalaxmi

Author S. P. P

Author R. A. M

Author P. S

Author D. P

Author G. E

Abstract Deepfakes entails obscene videos in which a face can be changed with someone else's utilizing neural networks. Deepfakes are a public problem, thus developing methods to detect them is critical. With deepfake and human-level control, deep learning has already utilized to address complicated issues. Deepfake is a new branch of AI technology, which in that one's face is superimposed on another's face, which is popular on social media. Deep learning techniques were also employed in the development of software that affects national security, democracy and privacy. One of the most current applications that use deep learning is deepfakes. Algorithms for deepfake may generate fake photos and movies which are indistinguishable from real ones. As a result, the development of technology capable of automatically detecting and assessing the reliability of digital video is critical. This research describes algorithms for detecting deepfakes. By reviewing the influence of deepfakes and deepfake recognition systems, this work enables the creation of new and so many effective methodologies to cope with increasingly complex deepfakes. InceptionResNetV2 architecture in Convolutional Neural Networks (CNN) is utilized in this comparative study to distinguish real and deepfake images.

Date 23-25 Feb. 2023

DOI 10.1109/ICCMC56507.2023.10083584

Extra Journal Abbreviation: 2023 7th International Conference on Computing Methodologies and Communication (ICCMC)

Pages 580-586

Proceedings Title 2023 7th International Conference on Computing Methodologies and Communication (ICCMC)

Conference Name 2023 7th International Conference on Computing Methodologies and Communication (ICCMC)

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Tags:

Deep learning, Privacy, Social networking (online), Software algorithms, Neural networks, CNN, Deep Learning, Face recognition, Deepfake, Deepfakes, Inception ResNet-V2

Deployment of Compressed MobileNet V3 on iMX RT 1060

Item Type Conference Paper

Author S. P. K. Prasad

Author M. El-Sharkawy

Abstract Deep Neural Networks (DNN) are prominent in most applications today. From self-driving cars, sentiment analysis, surveillance systems, and robotics, they have been used extensively. Among DNNs, Convolutional Neural Networks (CNN) have achieved massive success in computer vision applications as the human visual system inspires their architecture. However, striving to achieve higher accuracies, CNN complexity, parameters, and layers were increased, which led to a drastic surge in their size, making their deployment challenging. Over the years, many researchers have proposed various techniques to alleviate this issue—one of them being Design Space Exploration (DSE) to minimize size and computation with little compromise to accuracy. MobileNet V3 is one such architecture designed to achieve good accuracy while being mindful of resources. It produces an accuracy of 88.93% on CIFAR-10 with a size of 15.3MB. This paper further reduces its size to 2.3MB while boosting its accuracy to 89.13% using DSE techniques. It is then deployed into NXP's i.MX RT1060 Advanced Driver Assistance System (ADAS) platform.

Date 21-24 April 2021

DOI 10.1109/IEMTRONICS52119.2021.9422512

Extra Journal Abbreviation: 2021 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS)

Pages 1-4

Proceedings Title 2021 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS)

Conference Name 2021 IEEE International IOT, Electronics and Mechatronics Conference (IEMTRONICS)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Sentiment analysis, Computer architecture, Neural networks, Surveillance, Conferences, Mechatronics, Visual systems

Design of a Novel Priority Mux Based Indoor Autonomous Surveillance Robot

Item Type Conference Paper

Author S. Mythri

Author N. G. Bharadwaj

Author N. Pai

Author Ashwini

Author N. S. Soraganvi

Author B. N. Krupa

Abstract Bank robberies have exacerbated with time. It is a daunting task to monitor the banks for human-based security, considering long hours. Autonomous surveillance robots, in such situations, are beneficial to patrol a particular environment by collecting data with an electronic payload. DRISHTI, the proposed robot model designed using Robot Operating System (ROS), includes facial recognition to raise an alert if any suspect is detected, and human tracking is implemented during night patrolling. An Intrusion Alarm Panel (IAP) is employed using external sensors to detect any unethical breaking of the vault in addition to the anti-theft feature. The robot surveillance is aided by autonomous patrolling using Cartographer-based Simultaneous Localization and Mapping (SLAM). The accuracy of the generated map of the prototype is found to be 95.31% with respect to the corresponding ground truth. The design of a novel priority mux is proposed to integrate the surveillance functionalities with the navigation node. The alert velocity command node is made to override the navigation velocity commands with a 10 times priority ratio using the novel priority mux with a latency of less than one millisecond.

Date 17-17 June 2023

DOI 10.1109/I2CACIS57635.2023.10193057

Extra Journal Abbreviation: 2023 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)

Pages 292-297

Proceedings Title 2023 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)

Conference Name 2023 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Face recognition, Navigation, Heuristic algorithms, Autonomous Navigation, ROS, Surveillance, Simultaneous localization and mapping, IAP, Sockets, Vibrations

Developing a Prototype for Autonomous Drones

Item Type Conference Paper

Author S. Kalta

Author G. Bawa

Author G. Singh

Author H. Chauhan

Author A. Bag

Abstract Autonomous and advanced robotic systems have grown in popularity in recent years due to their wide range of applications, particularly in manufacturing, logistics, and other industries that require repetitive or precise work. These systems have proven to improve efficiency, productivity, and cost savings while

prioritizing worker safety by eliminating the need for humans to perform repetitive or hazardous tasks. This study provides an overview of the design and development of an autonomous drone system, with an emphasis on key features and capabilities of the system. This research study proposes a novel design and development of an autonomous drone system capable of performing predefined missions with precision and dependability without the need for human intervention. The prototype makes use of several advanced technologies, such as onboard obstacle avoidance sensors and weather monitoring sensors. Various algorithms are implemented to feed data from sensors to the flight controller to control the drone's flight path, navigate waypoints, and avoid obstacles autonomously. The research outcomes will include successful autonomous flight demonstrations, highlighting the drone's ability to complete complex missions. In conclusion, this paper focuses on the design and development of an autonomous drone system, demonstrating its potential to revolutionize multiple industries while also improving efficiency, safety, and sustainability when used responsibly and ethically. Refer to additional topics for a more in-depth discussion of autonomous drones.

Date 13-14 March 2024

DOI 10.1109/ICSADL61749.2024.00122

Extra Journal Abbreviation: 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

Pages 702-708

Proceedings Title 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

Conference Name 2024 3rd International Conference on Sentiment Analysis and Deep Learning (ICSADL)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Task analysis, Sensors, Prototypes, Monitoring, Robot sensing systems, Autonomous Robot, Flight Controller, Flight path, Obstacle Avoidance Sensors, Predefined Missions, Service robots, Weather Monitoring

Differentially Private Tensor Deep Computation for Cyber–Physical–Social Systems

Item Type Journal Article

Author N. J. Gati

Author L. T. Yang

Author J. Feng

Author S. Zhang

Author Z. Ren

Abstract In the recent past, deep learning has received remarkable acceptance in real-world applications. Social computing expands the existing notion of cyber space and physical space to a more advance cyber-physical-social system (CPSS). Therefore, deep learning provides a propitious technique for accurate mining of information

from CPSS, thus facilitates CPSS to offer services of exceptional quality efficiently. However, most of the current deep learning methods are struggling to keep up with the ever-increasing heterogeneous and highly nonlinear dissemination of data. Furthermore, the advancement of deep learning presents privacy concerns. This article proposes a deep private tensor autoencoder (dPTAE), where tensors are used for data representation, and differential privacy guarantees strong privacy. The core idea of our work is to enforce differential privacy through noise injection into the objective functions instead of the results they produce. In addition, the proposed method preserves the privacy of information shared amongst CPSS in smart environments. We applied dPTAE on three representative data sets. Rigorous experimental evaluations and theoretical analysis demonstrate that dPTAE is significantly effective and efficient.

Date Feb. 2021

DOI 10.1109/TCSS.2020.3004148

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Issue 1

Journal Abbr IEEE Transactions on Computational Social Systems

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Tags:

Deep learning, Privacy, privacy, Differential privacy, deep learning, Training, Computer science, cyber-physical-social systems (CPSS), deep computation, social computing, tensor autoencoder (TAE), Tensors, ϵ -differential privacy

Digital Forensics Challenges to Big Data in the Cloud

Item Type Conference Paper

Author X. Feng

Author Y. Zhao

Abstract As a new research area, Digital Forensics is a subject in a rapidly developing society. Cyber Security for Big Data in the Cloud is getting more attention than ever. A computing breach requires digital forensics to seize digital evidence to determine who is responsible and what has been done maliciously and the possible further consequences. In particular, for Big Data attack cases, Digital Forensics is facing even more challenge for earlier digital breach investigations. For the PPI (Protection of Personal Information) a GDPR (General Data Protection Regulation) law has been launched to be implemented from the 25th May 2018. This compulsory regulation will have an important impact on healthcare PPI in the cloud (ICO, 2017, Deloitte, 2014). Nowadays, Big Data with the characteristics of three “V”s (Volume, Velocity, and Variety), are either synchronized with the Cloud, or stored in the Cloud, in order to solve the storage capacity and so on problems, which made Digital Forensics investigation even more difficult. The Big Data Digital Forensics issue for the Cloud is difficult. One

of them is the need to identify which physical devices have been compromised. Data are distributed in the Cloud, so the customer or digital forensics practitioner cannot have full access control like the traditional investigation does. Smart City are making use of ICT (information communications technology) to collect, detect, analyze and integrate the key information data of core systems in running the cities. Meanwhile, the Control Centre is making intelligent responses to different requirements that include daily livelihood, PPI security, environmental protection, public safety, industrial and commercial activities and city services. The Smart City healthcare Big Data are collected and gathered by the IoT (Internet of Things) (Liu, 2014, Qi, 2016) and applying GDPR prevent Cyberstalking and Cybercrimes. This paper summerises our review on the trends of Digital Forensics used for Big Data. The evidence acquisition challenge is discussed. A case study of a Smart City project with IoT services collecting Big Data which are stored in the Cloud computing environment is represented. The techniques can be generalised to other Big Data in the Cloud environment.

Date 21-23 June 2017

DOI 10.1109/iThings-GreenCom-CPSCCom-SmartData.2017.132

Extra Journal Abbreviation: 2017 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCCom) and IEEE Smart Data (SmartData)

Pages 858-862

Proceedings Title 2017 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCCom) and IEEE Smart Data (SmartData)

Conference Name 2017 IEEE International Conference on Internet of Things (iThings) and IEEE Green Computing and Communications (GreenCom) and IEEE Cyber, Physical and Social Computing (CPSCCom) and IEEE Smart Data (SmartData)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Big data, Green computing, IoT, Cloud Computing, Internet of Things, Conferences, Cyber Security and Personally Identifiable Information (PII) GDPR and Protection of Personally Information (PPI) against Cyberstalking, Smart City system, Social computing

Dynamic Service Function Chain (SFC) Deployment for Autonomous Intelligent Systems

Item Type Journal Article

Author X. Kong

Author L. Peng

Author S. Rahim

Abstract Autonomous Intelligent Systems (AIS), such as autonomous vehicles, smart drones, and robotic swarms, are becoming increasingly prevalent in next-generation intelligent infrastructures. However, AIS environments are characterized by highly dynamic network conditions, severe device heterogeneity, strict end-to-end latency

requirements, and strong privacy constraints due to distributed multimodal sensing data. These challenges render traditional Service Function Chain (SFC) deployment approaches, especially centralized or slow-adapting methods, unsuitable for real-world AIS scenarios. To address these challenges, we propose MetaFedDRL, a meta-federated reinforcement learning framework designed for fast, privacy-preserving, and communication-efficient SFC deployment in large-scale AIS networks. MetaFedDRL integrates Model-Agnostic Meta-Learning (MAML), Proximal Policy Optimization (PPO), and Asynchronous Federated Learning (AFL) to learn a generalizable policy initialization across diverse environments. This enables rapid adaptation to new conditions with only a few local gradient updates, significantly reducing convergence time. The framework exchanges only lightweight gradient information, thereby preserving data privacy and minimizing communication overhead. Extensive simulations under dynamic and heterogeneous AIS settings show that MetaFedDRL achieves higher SFC acceptance rates, lower end-to-end latency, reduced average resource consumption, and improved deployment fairness compared to state-of-the-art baselines.

Date Nov. 2025

DOI 10.1109/TCE.2025.3612966

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Publication IEEE Transactions on Consumer Electronics

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Tags:

Data privacy, Artificial intelligence, Drones, Costs, Bandwidth, asynchronous federated learning, autonomous intelligent systems, low-latency deployment, model-agnostic meta-learning, privacy preservation, Service function chain deployment, Computational modeling, Delays, Resource management, Training, Optimization

Dynamic Traffic Optimization Through Cloud-Enabled Big Data Analytics and Machine Learning for Enhanced Urban Mobility

Item Type Conference Paper

Author E. Anvesh

Author D. Brindha

Author C. H. Kumar

Abstract Addressing the critical problem of road traffic congestion that robs people of time, ruins the environment, and eats into the purse to the detriment of urban mobility, this paper. The research is relevant as urban congestion remains a serious issue currently, and for which new traffic management solutions for more efficient traffic management are needed. The intelligent traffic control system, developed via the integration of cloud computing, big data analytics and machine

learning (using the RandomForest algorithm for predictive analysis) is the proposed solution. From sensors, GPS devices, and social networks, the system collects real time traffic data, and retimes and reroutes vehicles in accordance with current traffic conditions. Results show that the system can: increase safety in urban environments, reduce congestion, and improve traffic flow. The results indicated perfect classification accuracy by the RandomForest algorithm with accurate real time traffic prediction and effective management of traffic. It is concluded that cloud infrastructure integration results in scalability and flexibility that makes the infrastructure capable of handling large datasets. Some potential applications include a wide scale deployment in smart cities and future research can further enhance traffic safety and reduce travel time by further improving predictive models and using AI based decision making.

Date 4-6 April 2025

DOI 10.1109/ICCSAI64074.2025.11063940

Extra Journal Abbreviation: 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI)

Volume 3

Pages 469-474

Proceedings Title 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI)

Conference Name 2025 3rd International Conference on Communication, Security, and Artificial Intelligence (ICCSAI)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Transportation, machine learning, Machine learning, Traffic control, Traffic congestion, Big Data, traffic congestion, traffic management, cloud computing, Cloud computing, Vehicle dynamics, Real-time systems, Predictive models, predictive modeling, Accuracy, scalability, adaptive traffic control, big data analytics, dynamic traffic control, RandomForest, real-time data, traffic optimization, urban mobility, urban transportation, vehicle simulation

Enhancing IoT Network Security through Adaptive Curriculum Learning and XAI

Item Type Conference Paper

Author S. Narkedimilli

Author S. Makam

Author A. V. Sriram

Author S. Prashanth Mallellu

Author M. Sathvik

Author R. V. Prasad

Abstract To address the critical need for secure IoT networks, this study presents a scalable and lightweight Curriculum Learning framework enhanced with Explainable AI (XAI) techniques, like LIME, to ensure transparency and adaptability. The

proposed model employs a novel neural network architecture utilized at every stage of Curriculum Learning to efficiently capture and focus on both short- and long-term temporal dependencies, improve learning stability, and enhance accuracy while remaining lightweight and robust against noise in sequential IoT data. Robustness is achieved through staged learning, where the model iteratively refines itself by removing low-relevance features and optimizing performance. The workflow includes edge-optimized quantization and pruning to ensure portability that could easily be deployed in edge IoT devices. An ensemble model incorporating Random Forest, XGBoost, and the staged learning base further enhances generalization. The results demonstrate 98% accuracy on CIC-IoV-2024 and CIC-APT-IIoT-2024 datasets and 97% on EDGE-IIoT, establishing this framework as a robust, transparent, and high-performance solution for IoT network security.

Date 10-12 Nov. 2025

DOI 10.1109/FNWF66845.2025.11317635

Extra Journal Abbreviation: 2025 IEEE Future Networks World Forum (FNWF)

ISBN 2770-7679

Pages 1-6

Proceedings Title 2025 IEEE Future Networks World Forum (FNWF)

Conference Name 2025 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Network security, Neural networks, Internet of Things, Explainable AI, Robustness, Data models, Accuracy, Training, Adaptation models, Curriculum Learning, Explainable Artificial Intelligence (XAI), Intrusion Detection System, IoT Network Security, Thermal stability

Envisioning Cybersecurity Analytics for the Internet of Things

Item Type Conference Paper

Author J. S. Chavis

Author D. P. Syed

Abstract The Internet of Things represents a rethinking of how we currently envision networks. It involves increases in scale of an order of magnitude over the next decade, increasingly distributed and virtualized architectures, and an ever-changing perimeter. Cybersecurity analytic capability will need to adapt to meet the demands of this evolving environment. In this paper the authors assess the ways in which cybersecurity analytics will need to evolve and identify the potential role of Government in promoting needed changes. This paper provides a brief summary of that effort.

Date 10-12 Sept. 2020

DOI 10.1109/5GWF49715.2020.9221018

Extra Journal Abbreviation: 2020 IEEE 3rd 5G World Forum (5GWF)

Pages 193-198

Proceedings Title 2020 IEEE 3rd 5G World Forum (5GWF)

Conference Name 2020 IEEE 3rd 5G World Forum (5GWF)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Visualization, IoT, Internet of Things, Quality of service, Heuristic algorithms, Telecommunication traffic, Market research, Government, Cybersecurity Analytics

Evaluating Very Large-Scale Integration (VLSI) for Deep Learning: Embedded Systems for Processing Images and Videos in Real Time

Item Type Conference Paper

Author A. J

Author S. D

Author K. Nivetha

Author Z. Azzam

Author H. P. Thethi

Author M. Dinesh

Abstract The need for efficient and high-performance computing systems have been driven by the growing demand for real-time image and video processing in applications such as autonomous vehicles, surveillance, augmented reality, and medical diagnostics. The need to generate these growing amounts of data has resulted in a new age where VLSI technology is regarded as a critical enabler for achieving the required computational power and energy efficiency in deep-learning-based applications. This paper performs a survey on how VLSI circuits can be integrated to perform deep learning tasks specifically targeting embedded systems that handle real-time processing of images and videos. VLSI plays a pivotal role in significantly improving the performance of executing the DNN operation execution, whether it is in the form of a CNN-based architecture or RNN architecture, while also addressing the trade-offs concerning speed, power consumption, and cost in the same process. An in-depth comparison of existing VLSI design paradigms and hardware architectures for real-time deep learning, focusing on efficient processing of large datasets on embedded systems. FPGAs are hardware platforms that can be programmed to implement a wide range of digital circuits, providing flexibility for custom neural networks, while ASICs are optimized for specific network architectures but lack reprogramability. Real-time video processing is discussed along with application examples of VLSI-based systems in different real applications, such as facial recognition, object detection, and scene segmentation. Due to the inherent similarities with current processing systems, the evaluation also includes a comparison against their traditional processing counterparts, demonstrating the transformative advancements in throughput and energy consumption achieved via VLSI. Lastly, their paper goes into detail on the hardware/software co-design, shortcomings of the memory bandwidth, and the parallelism in the architecture of the deep learning network. More specifically, the study points to the growing demand for significantly parallelized computing systems to handle the terabytes of data produced by image and video sensors. Furthermore, the analysis includes a discussion on the influence of deep learning model optimization methods like pruning,

quantization, and hardware-aware training on VLSI hardware efficiency.

Date 3-4 Sept. 2025

DOI 10.1109/ICEC2NT65402.2025.11379985

Extra Journal Abbreviation: 2025 International Conference on Electronics and Computing, Communication Networking Automation Technologies (ICEC2NT)

Pages 1-6

Proceedings Title 2025 International Conference on Electronics and Computing, Communication Networking Automation Technologies (ICEC2NT)

Conference Name 2025 International Conference on Electronics and Computing, Communication Networking Automation Technologies (ICEC2NT)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Deep learning, Computer architecture, Computational efficiency, Embedded systems, Real-time systems, Computational modeling, Videos, Application-Specific Integrated Circuits (ASICs), Embedded Systems, Field programmable gate arrays, Field-Programmable Gate Arrays (FPGAs), Hardware Acceleration, Image and Video Processing, Integrated circuit modeling, Real-Time Processing, Very large scale integration

Explainable Forecasts of Disruptive Events using Recurrent Neural Networks

Item Type Conference Paper

Author A. L. Buczak

Author B. D. Baugher

Author A. J. Berlier

Author K. E. Scharfstein

Author C. S. Martin

Abstract This paper describes the Crystal Cube method we developed for forecasting disruptive events around the world, specifically Irregular Leadership Change. Crystal Cube uses a Recurrent Neural Network (RNN) with Long-Short Term Memory (LSTM) units for forecasting. In this paper special emphasis is put on explanations of the network forecasts. We are using SHapley Additive exPlanations (SHAP) for individual forecast explanations and we are aggregating the explanations separately for True Positives, False Positives, True Negatives, and False Negatives. The method can be extended to Deep Reinforcement Learning models for self-driving cars or unmanned fighter jets.

Date 22-24 March 2022

DOI 10.1109/ICAA52185.2022.00017

Extra Journal Abbreviation: 2022 IEEE International Conference on Assured Autonomy (ICAA)

Pages 64-73

Proceedings Title 2022 IEEE International Conference on Assured Autonomy (ICAA)

Conference Name 2022 IEEE International Conference on Assured Autonomy (ICAA)

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Tags:

Deep learning, Reinforcement learning, Recurrent neural networks, AI safety, Leadership, Conferences, Predictive models, RNN, acceptance of machine learning models, Crystals, disruptive event, explainable deep learning, forecasting, interpretable machine learning, LSTM, resilient, SHAP

Exploring Graduate Student Views on Robotics Policy Relevant Information

Item Type Conference Paper

Author S. Balali

Author J. Parham-Mocello

Author B. Steel

Author M. D. Jones

Author W. Billah

Abstract This full research paper explores the perspectives of Computer Science (CS) and Public Policy (PPOL) graduate students on the presence of embodied AI systems (i.e., AI-powered robots) in public spaces. Robots in public spaces pose unique challenges due to their embodiment, emergence, and social valence, raising societal, ethical, and regulatory concerns. These challenges disrupt established legal norms, requiring nuanced policies. As AI becomes more integrated into daily life, public concern about its societal impact grows, emphasizing the need for robust technology policies and ethical education. Research shows that public perceptions of AI are often more negative than views on algorithms, associating AI with fear and uncertainty. While studies address public perceptions of AI, little research focuses on how future technologists and policymakers perceive embodied AI and its challenges. To address this, we used the Narrative Policy Framework (NPF) to investigate CS and PPOL students' views on concerning robots and related policy issues. Through semi-structured interviews with three CS and four PPOL graduate students, we identified five categories of concerning robots, with self-driving cars and humanoid robots being the most prominent. We examined 10 policy challenges, 16 policy solutions, and the roles of key stakeholders. These findings offer insights into interdisciplinary education, AI policy development, and robotics literacy, fostering collaboration between technical and societal fields and preparing students to develop ethical AI technologies and informed public policies.

Date 2-5 Nov. 2025

DOI 10.1109/FIE63693.2025.11328351

Extra Journal Abbreviation: 2025 IEEE Frontiers in Education Conference (FIE)

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Proceedings Title 2025 IEEE Frontiers in Education Conference (FIE)

Conference Name 2025 IEEE Frontiers in Education Conference (FIE)

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Tags:

Artificial intelligence, Public policy, Education, Robots, Interviews, Stakeholders, Law, Ethics, Computer science, Graduate Students' Robotics Policy Understanding, Humanoid robots, Narrative Policy Framework (NPF), Semi-structured Interviews

Exploring Inherent Backdoors in Deep Learning Models

Item Type Conference Paper

Author G. Tao

Author S. Cheng

Author Z. Wang

Author S. Ma

Author S. An

Author Y. Liu

Author G. Shen

Author Z. Zhang

Author Y. Mao

Author X. Zhang

Abstract Deep learning has been widely integrated into a variety of real-world systems, such as facial recognition and autonomous driving. However, recent studies demonstrate that deep learning models are vulnerable to backdoor attacks. These attacks inject a backdoor trigger into input samples, causing them to be misclassified to an attacker-chosen target output. Existing backdoor attacks are typically carried out by poisoning the training data or modifying model weight parameters. In this paper, we show that backdoor attacks can be realized without poisoning the data or model. Backdoors can be widely identified in normally trained clean models, which we call inherent backdoors. To find such backdoor vulnerabilities, we summarize and categorize 20 existing injected backdoor attacks and leverage them to guide the search for inherent backdoors. Specifically, we define backdoor vulnerabilities based on four important properties and characterize them according to how they manipulate the input and constrain the changes. We conduct a systematic study on 54 pre-trained legitimate models downloaded from trusted sources and find 315 inherent backdoors in these models, covering all different categories. We also study the potential causes for inherent backdoors and how to defend against them.

Date 9-13 Dec. 2024

DOI 10.1109/ACSAC63791.2024.00078

Extra Journal Abbreviation: 2024 Annual Computer Security Applications Conference (ACSAC)

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Tags:

Autonomous vehicles, Deep learning, Face recognition, Computer security, Computational modeling, Data models, Systematics, Training data

Exposing Adaptive Cruise Control in Advanced Driving Assistance Systems

Item Type Conference Paper

Author M. A. Hoque

Author R. Hasan

Abstract Advanced Driving Assistance Systems (ADAS) are expected to improve the urban driving experience by reducing accidents and enabling safe driving. Such systems provide different features to automate driving and ease the driver's tasks. ADAS features highly depend on the data received from in-vehicle networks, such as Controller Area Network (CAN) bus. CAN play an important role in in-vehicle communication by enabling efficient data exchange among electronic control units (ECUs). However, the CAN bus is vulnerable to different attack strategies, which can manipulate CAN data and cause fatal consequences on ADAS features. This paper shows how CAN bus data can be exploited to expose an industry-grade advanced driving assistance system's Adaptive Cruise Control (ACC) feature. The attack technique demonstrates the requirements of timely human intervention or automatic recovery techniques. We also provide a detection mechanism based on the high-definition vector map and Global Positioning System device installed in the vehicle.

Date 26 Oct.-11 Nov. 2022

DOI 10.1109/WF-IoT54382.2022.10152232

Extra Journal Abbreviation: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

Pages 1-6

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Conference Name 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

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Tags:

Accidents, Task analysis, security, Internet of Things, Global Positioning System, adaptive cruise control, Adaptive systems, advanced driving assistance systems, controller area network, Cruise control, HD map

Facilitating Citizen Engagement in Smart City Mobility

Item Type Conference Paper

Author A. R. Patel

Author R. Ahuja

Author M. Roscia

Abstract As cities around the world embrace smart mobility solutions to address transportation challenges, it is crucial to engage citizens in the development and implementation of these initiatives. Smart city mobility encompasses a range of technologies and strategies aimed at improving transportation efficiency, reducing congestion and emissions, and enhancing accessibility for all residents. However, without meaningful citizen engagement, these solutions run the risk of failing to meet the diverse needs and priorities of the community. This paper explores various approaches to facilitating effective citizen participation in smart city mobility planning and decision-making processes. It examines best practices for gathering citizen input through public forums, surveys, and digital platforms, as well as strategies for incorporating this feedback into policy and project development. The paper also discusses the importance of transparency, education, and capacity-building to empower citizens to make informed contributions. By fostering a collaborative and inclusive approach, cities can ensure that smart mobility solutions are tailored to the unique contexts and aspirations of their communities, ultimately leading to more sustainable, equitable, and citizen-centric urban transportation systems.

Date 31 July-3 Aug. 2024

DOI 10.1109/SEFET61574.2024.10717973

Extra Journal Abbreviation: 2024 IEEE 4th International Conference on Sustainable Energy and Future Electric Transportation (SEFET)

Pages 1-6

Proceedings Title 2024 IEEE 4th International Conference on Sustainable Energy and Future Electric Transportation (SEFET)

Conference Name 2024 IEEE 4th International Conference on Sustainable Energy and Future Electric Transportation (SEFET)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Transportation, Decision making, Sustainable development, Technological innovation, Smart cities, Surveys, Collaboration, Socioeconomics, Resilience, Planning, Mobility, Citizen participation, Urban mobility development

Flooded area detection using UAV images

Item Type Conference Paper

Author A. Sumalan

Author D. Popescu

Author L. Ichim

Abstract In this paper, we propose an unsupervised technique for the detection of flooded areas using images acquired by a fixed-wing unmanned aerial vehicle (UAV). The proposed methodology for the detection of the flooded areas considers the development of an algorithm based on k means clustering and texture analysis using co-occurrence matrix.

Date 22-23 Nov. 2016

DOI 10.1109/TELFOR.2016.7818798

Extra Journal Abbreviation: 2016 24th Telecommunications Forum (TELFOR)
Pages 1-4
Proceedings Title 2016 24th Telecommunications Forum (TELFOR)
Conference Name 2016 24th Telecommunications Forum (TELFOR)
Date Added 20/09/2026, 7:47:03 pm
Modified 20/09/2026, 7:47:03 pm

Tags:

Feature extraction, Remote sensing, Clustering algorithms, Cameras, Unmanned aerial vehicles, flood evaluation, Floods, k means clustering, texture analysis, UAV surveillance

Flying Communication Server in case of a Largescale Disaster

Item Type Conference Paper
Author T. Kobayashi
Author H. Matsuoka
Author S. Betsumiya
Abstract We have developed a Flying Communication Server, which is a drone, unmanned aerial vehicle (UAV), equipped with a single board computer implemented in a Wi-Fi base station, a Web server and a WebRTC server. With this system, even in situations in which the cellular telephony network/public wireless network have been collapsed due to a large-scale disaster, users who have their own private smart devices such as smart phones will be able to share the video of an affected area from the sky, exchange the text chat asynchronously, and execute the real-time video communication. In order to make it possible, we proposed three communication layers architecture based on Wi-Fi standard technologies and Web technologies. In this study, we have checked a range of connectivity of the Flying Communication Server such as Wi-Fi distance, frame rate to distribute video stream, and concurrent connection user numbers. We have also evaluated the text chat and the real-time video communication availability. Then, we have confirmed the effectiveness of the Flying Communication Server concerning to information sharing and asynchronous/realtime communication in case of a large-scale disaster.
Date 10-14 June 2016
DOI 10.1109/COMPSAC.2016.117
Extra Journal Abbreviation: 2016 IEEE 40th Annual Computer Software and Applications Conference (COMPSAC)
Volume 2
ISBN 0730-3157
Pages 571-576
Proceedings Title 2016 IEEE 40th Annual Computer Software and Applications Conference (COMPSAC)
Conference Name 2016 IEEE 40th Annual Computer Software and Applications Conference (COMPSAC)
Date Added 20/09/2026, 7:47:03 pm
Modified 20/09/2026, 7:47:03 pm

Tags:

UAV, Smart phones, Cameras, Smart devices, Servers, Communication Server, Computers, Disaster, Drone, IEEE 802.11 Standard, Single Board Computer, Streaming media, WebRTC, WebSocket

Focusing on the Ethical Challenges of Data Breaches and Applications

Item Type Conference Paper

Author K. Joisten

Author N. Thiemer

Author T. Renner

Author A. Janssen

Author A. Scheffler

Abstract Ethical challenges of the human lifeworld that are caused by data breaches and applications are steadily increasing. Therefore, a new ethical concept must be brought into focus: Technoethics for Emerging Digital Systems (TEDS). TEDS is presented as an integrative and innovative approach committed to an interdisciplinary perspective. Thereby, TEDS reflects all social areas of the human lifeworld in their ethical scope. With recourse to phenomenological methods, TEDS helps to address ethical implications which arise from deep interference of autonomous systems with the human lifeworld. The meaning of intentional structures and the problem of appresentations in the phenomenological sense still represents a blank gap in the current ethical discourse on the problem of data breaches. The findings provide methods for dealing with ethical challenges and explain the problem area of an appropriate technoethical use in the lifeworld. In this way, problems can already be avoided in the development process of artificial intelligence systems and their applications by specifically searching for blind spots in a technical and ethical manner. Furthermore, this approach helps to assure a technoethical use of autonomous systems in an appropriate way and ultimately leads to a limitation of damages – which may occur in case of malfunctions and data breaches of artificial intelligence systems – in the lifeworld of humans. Our contribution is to introduce TEDS as a new ethical concept that has not existed before. This new concept focuses on the application of phenomenological methods to detect ethical errors in digital systems.

Date 22-24 March 2022

DOI 10.1109/ICAA52185.2022.00018

Extra Journal Abbreviation: 2022 IEEE International Conference on Assured Autonomy (ICAA)

Pages 74-82

Proceedings Title 2022 IEEE International Conference on Assured Autonomy (ICAA)

Conference Name 2022 IEEE International Conference on Assured Autonomy (ICAA)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Data privacy, AI, privacy, security, ethics, Ethics, autonomous systems, Digital systems, Autonomous systems, Interference, Focusing, application, Data breach, phenomenology, physical world/lifeworld, Technoethics

Identification and analysis of charging factors in M2M communications

Item Type	Conference Paper
Author	F. J. Lin
Author	W. Tsai
Author	E. Cerritos
Author	B. -T. Lin
Author	W. -H. Hu
Abstract	The widespread application of M2M communications is obvious. There are already many useful services enabled by M2M communications such as video surveillance, smart grid, smart home, connected vehicle, industry automation and smart city. These services are different from traditional Internet services such as e-mail, web browsing and video watching. As a result, we need to understand new charging factors beyond those defined in the past such as amount of data transfer, number of short messages and duration of voice communication. In this paper, we identify new charging factors for M2M communications and analyze their differences with traditional voice and Internet networks. Based on the result of our analysis, new charging models specific to M2M communications are proposed to enable smart data pricing that creates a win-win situation for operators and users.
Date	14-16 Dec. 2015
DOI	10.1109/WF-IoT.2015.7389045
Extra	Journal Abbreviation: 2015 IEEE 2nd World Forum on Internet of Things (WF-IoT)
Pages	160-165
Proceedings Title	2015 IEEE 2nd World Forum on Internet of Things (WF-IoT)
Conference Name	2015 IEEE 2nd World Forum on Internet of Things (WF-IoT)
Date Added	20/09/2026, 7:47:04 pm
Modified	20/09/2026, 7:47:04 pm

Tags:

Computer architecture, Quality of service, Data transfer, Sensors, Telecommunications, Logic gates, ETSI, M2M charging factors, M2M charging models, M2M communications

INGR Roadmap Edge Services and Automation Chapter

Item Type	Conference Paper
Author	M. Patwary
Author	P. Ramchandran
Author	S. Tibrewala

Author T. K. Lala

Author F. Kautz

Author E. Coronado

Author R. Riggio

Author S. Ganugapati

Author S. Ranganathan

Author L. Liu

Author G. Giambene

Author Z. Sun

Author M. Borst

Author B. Kolaz

Author A. Dutta

Author C. -M. Chen

Author M. Bhandaru

Abstract This third edition (2023) continues to reflect the Edge service roadmap and journey with evolving key drivers and developments. It replaces the two earlier versions. 5G deployments have reached critical markets and are fast evolving towards 6G promising higher performance and near sub-millisecond latency. Higher automation in use cases fueled by AI (e.g. like ChatGPT) is expected to be the core of future applications and services. The service providers have started re-focusing on use cases that can leverage the new and improved capabilities for better average revenue per unit (ARPU) for both consumers and enterprises. Some of the edge relevant Service oriented use cases are based on key constraints (low latency, high throughput, low jitter) ▪Industrial Internet of Things (IIoT) Industry 4.0 ▪Vehicle to anything - V2X (autonomous vehicle / intelligent transportation / traveling edge) ▪Telehealth / telemedicine / remote diagnostics ▪Content delivery (with caching, real time, rich media internet applications) ▪Ad hoc, temporary or on as needed mission specific edge services (emergency, ad hoc major events, DoD combat mission, and more) ▪Enabling sustainable development edge applications. Key elements of the edge-inspired infrastructure to support such services may be noted as follows: ▪Enabling edge with AIML, e.g., OpenAI / *ChatGPT ▪Radio-based Multi-Access Edge (MEC) to provide different services ▪Real-time / near real time enabled radio control, management, and xApps based on O-RAN ▪Constraint-based edge infrastructure to optimize power, form factor, and updates to NFV-SDN with the IoT middleware standards OneM2M from ETSI to cover IoT applications ▪The focus on data privacy, security, data “localization and analytics” ▪Intelligent edge microservices and applications deployed using newer infrastructure components from Semiconductor innovations e.g. DPU, IPU, etc. ▪MEC and 5G integration to support location-based edge platform API adaption ▪Small cells and private B5G networks We classify platform aspects with Edge Platform Framework (EPF) and Services with Edge Service Framework (ESF) updates and overall aspects as Edge Service and Platform Framework (ES&PF) for an integrated view.

Date 13-15 Nov. 2023

DOI 10.1109/FNWF58287.2023.10520517

Extra Journal Abbreviation: 2023 IEEE Future Networks World Forum (FNWF)

ISBN 2770-7679

Pages 1-68

Proceedings Title 2023 IEEE Future Networks World Forum (FNWF)

Conference Name 2023 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Vehicles, Automation, Sustainable development, Technological innovation, Vehicle-to-everything, Telemedicine, Real-time systems, Standards, Throughput, US Department of Defense

INGR Roadmap Edge Services and Automation Chapter

Item Type Conference Paper**Author** M. Patwary**Author** P. Ramchandran**Author** S. Tibrewala**Author** T. Lala**Author** F. Kautz**Author** E. Coronado**Author** R. Riggio**Author** S. Ganugapati**Author** S. Ranganathan**Author** L. Liu

Abstract The Key drivers have certainly changed, and the 5G-hype has reached the cycle of disillusionment globally in 2021–2022. Communication Service Providers have started re-focusing on few use cases that can leverage the markets for better average revenue per unit (ARPU) for both consumers & Enterprises. The list is short includes 1. Service based use cases based on Service constraints (low latency, high throughput, low jitter) a. HOT (Industry 4.0) b. V2X (Autonomous Vehicle/ Intelligent Transportation Traveling Edge) c. Telehealth / Telemedicine Remote diagnostics d. Content delivery (with Caching. Realtime. Reach Media Internet Applications) e. Ad hoc. Temporary or as needed basis Specific mission Edge sendees (Emergency, ad hoc major events. DoD combat mission, and more) 2. Radio Based Multi-Access Edge (MEC) with different flavors a. Real Time / Near Real Time enabled Radio Control. Management and XApps based on O-RAN b. Constrained based Edge Infrastructure to optimize Power, Form Factor and Updates to NFV-SDN with OneM2M from ETSI to cover IoT Applications c. The focus on Data Privacy. Security. Data Localization & Analytics Big or Small d. Smart Edge with Containers and Compute & Network acceleration are emerging from Infra structure and Semiconductor players. We name item 1 in above as Edge Service Platform Framework (ESPF-2021) supplementary updates and item 2 in above as Edge Automation Platform Framework (EAPF-2021) updates for INGR.2021 efforts released here in 2022.

Date 10-14 Oct. 2022**DOI** 10.1109/FNWF55208.2022.00136**Extra** Journal Abbreviation: 2022 IEEE Future Networks World Forum (FNWF)**ISBN** 2770-7679**Pages** 1-49**Proceedings Title** 2022 IEEE Future Networks World Forum (FNWF)

Conference Name 2022 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Security, Vehicles, V2X, Automation, IoT, 5G, Telemedicine, Media, Real-time systems, Low latency communication, Throughput, US Department of Defense, EAPF, ENS, ESPF, IIOT, MEC, O-RAN, OneM2M, Radio control, XApps

Internet of Things Ecosystems: Unpacking Legal Relationships and Liabilities

Item Type Conference Paper

Author C. Millard

Author W. K. Hon

Author J. Singh

Abstract This paper provides a survey of key legal questions arising from the development of Internet of Things ecosystems. We analyse in particular the types of relationships and liabilities that are likely to emerge as Internet of Things ecosystems evolve and we consider implications for both organisations and individuals by reference to specific legal and regulatory considerations. In terms of jurisdictional scope, our main focus is the EU but we also provide examples from other parts of the world.

Date 4-7 April 2017

DOI 10.1109/IC2E.2017.46

Extra Journal Abbreviation: 2017 IEEE International Conference on Cloud Engineering (IC2E)

Pages 286-291

Proceedings Title 2017 IEEE International Conference on Cloud Engineering (IC2E)

Conference Name 2017 IEEE International Conference on Cloud Engineering (IC2E)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Ecosystems, policy, regulation, IoT, Cloud Computing, Internet of Things, technology, liability, Cloud computing, Law, Cloud, consumer protection, contracts, Contracts, cyber-risk, insurance, law

Internet of things prospect in Nigeria: Challenges and solutions

Item Type Conference Paper

Author O. J. Kunle

Author O. A. Olubunmi

Author S. Sani

Abstract This research analyzed key issues and challenges militating against IoT readiness of Nigeria and acceptance. Key issues considered include: Quality of Service, Quality of Experience, government will and policies, infrastructure deficits security issues, privacy etc. The solutions to the challenges were also clearly expatiated to include but not limited to good government policies, national orientation and creation of awareness, encouragement of development of infrastructure, collaborative regulation, enactment of IoT Framework, Nigerian active participation in international Telecommunications Union-T Study Group 20 on Internet of Things Smart Cities and communities and Study Group 13 based on Future Networks like 5G technology, development of relevant human capital and excellent skill sets. The work also dealt with the probable applications of IoT such as health applications, vehicular traffic management, remote monitoring, connected cars etc. The following areas are also identified as the gains of IoT deployment in Nigeria.

Date 7-10 Nov. 2017

DOI 10.1109/NIGERCON.2017.8281942

Extra Journal Abbreviation: 2017 IEEE 3rd International Conference on Electro-Technology for National Development (NIGERCON)

ISBN 2377-2697

Pages 736-745

Proceedings Title 2017 IEEE 3rd International Conference on Electro-Technology for National Development (NIGERCON)

Conference Name 2017 IEEE 3rd International Conference on Electro-Technology for National Development (NIGERCON)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Security, Internet of Things, Connected Vehicles, Smart Cities, Internet of Things (IoT), Government, International Telecommunications Union Internet of Everything (IoE), Netizen, Quality of Experience (QoE), Quality of Service (QoS), Radio Frequency Identity (RFID), Temperature sensors, Wearable

Intricate Plane of Adversarial Attacks in Sustainable Territory and the Perils faced Machine Intelligent Models

Item Type Conference Paper

Author K. S. Sai Susmitha

Author N. Srimlika

Author V. Sharma

Author M. Almusawi

Author S. Mishra

Abstract The issue of model security and reliability in Artificial Intelligence (AI) is a concern due to adversarial attacks. In order to tackle this issue, researchers have developed sustainable defense strategies, but certain challenges remain. These challenges involve transferability, higher computing costs, and adaptability. Striking a balance between accuracy and robustness is difficult, as defense

mechanisms often come with trade-offs between the two. Real-world situations demonstrate the practical implications of sustainable adversarial AI. For example, it improves the security of self-driving vehicles, enhances the accuracy of medical imaging diagnoses, and incorporates AI-driven defenses into network intrusion detection and phishing detection systems. It is crucial to consider ethical aspects throughout this process. Future trends in adversarial AI research for cybersecurity will involve ensemble defense mechanisms, adversarial learning from limited data, and hybrid attacks. By embracing the evolving landscape, researchers and practitioners can develop sustainable AI systems that are more secure and resilient, effectively countering adversarial threats.

Date 14-15 Nov. 2023

DOI 10.1109/ICTEASD57136.2023.10584909

Extra Journal Abbreviation: 2023 International Conference for Technological Engineering and its Applications in Sustainable Development (ICTEASD)

Pages 233-239

Proceedings Title 2023 International Conference for Technological Engineering and its Applications in Sustainable Development (ICTEASD)

Conference Name 2023 International Conference for Technological Engineering and its Applications in Sustainable Development (ICTEASD)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Cyber security, Safety, Network intrusion detection, Machine learning, Technological innovation, Deep Learning, Adversarial machine learning, Sustainability, Robustness, Accuracy, Adversarial AI, Generative Adversarial Networks(GANs), Phishing

Investigating eVTOL Passenger Preferences in South Korea: Safety, Vertiport Location, and Weight

Item Type Conference Paper

Author P. B. Appel

Author Y. Eun Song

Author A. Riener

Abstract As the industry advances towards the technical realization of passenger drones, there is a noticeable trend where the wishes and acceptance factors of future users are not given primary consideration. Although several companies are progressing with prototype development and market-entry preparations, ongoing research on the requirements for electric Vertical Take-Off and Landing (eVTOL) vehicles remains crucial. For such a novel form of aerial transportation, understanding passenger needs is key, as their acceptance and willingness to use these services will ultimately drive revenue for manufacturers and airlines. To this end, our study, involving 51 participants in South Korea, delved into three key areas of user acceptance: safety considerations of eVTOLs, preferred vertiport locations, and concerns about weight. The findings indicate that piloted passenger drones are acceptable to the public, particularly when they assure safety and offer sufficient luggage storage, among other factors. These insights are vital for the

engineering design and development of eVTOLs, highlighting opportunities for weight and energy savings by aligning with passenger preferences and requirements.

Date 2-5 June 2024

DOI 10.1109/IV55156.2024.10588875

Extra Journal Abbreviation: 2024 IEEE Intelligent Vehicles Symposium (IV)

ISBN 2642-7214

Pages 3269-3275

Proceedings Title 2024 IEEE Intelligent Vehicles Symposium (IV)

Conference Name 2024 IEEE Intelligent Vehicles Symposium (IV)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Transportation, Market research, Sociology, Prototypes, Intelligent vehicles, Technology transfer

Kernel Two Dimensional Subspace for Image Set Classification

Item Type Conference Paper

Author B. B. Gatto

Author W. S.S.

Author E. M. dos Santos

Abstract Object recognition on large-scale video has recently attracted considerable research interest due to the huge amount of data available on the Internet, surveillance systems, social media networks and autonomous vehicles. By representing large-scale videos as image sets, we can handle the complex data variations such as viewpoint, illumination, and pose. In this paper, we propose an efficient and robust method for image set recognition based on Kernel Orthogonal Mutual Subspace Method (KOMSM), where sets of images are expressed as nonlinear subspaces. In our method, we formulate the image sets as nonlinear 2D subspaces by applying K2D-PCA and variants of 2D-PCA. Comparing to KOMSM, the proposed method requires less memory resource since it inherits the computational advantages of 2D-PCA and variants. In addition, the subspaces produced by K2D-PCA preserves the spatial relation between image pixels, generating more informative subspaces than KOMSM. The introduced method has the advantage of representing the subspaces in a more compact manner, achieving lower time complexity, confirming the suitability of employing 2D-PCA and variants. These results have been revealed through comprehensive experimentation conducted on five publicly available datasets.

Date 6-8 Nov. 2016

DOI 10.1109/ICTAI.2016.0155

Extra Journal Abbreviation: 2016 IEEE 28th International Conference on Tools with Artificial Intelligence (ICTAI)

ISBN 2375-0197

Pages 1004-1011

Proceedings Title 2016 IEEE 28th International Conference on Tools with Artificial Intelligence (ICTAI)
Conference Name 2016 IEEE 28th International Conference on Tools with Artificial Intelligence (ICTAI)
Date Added 20/09/2026, 7:47:04 pm
Modified 20/09/2026, 7:47:04 pm

Tags:

Face recognition, Image recognition, Memory management, Covariance matrices, Image color analysis, Image set classification, Kernel, Kernel Methods, Principal component analysis, Two Dimensional PCA

Lane Change Intent Prediction Based on Multi-Channel CNN Considering Vehicle Time-Series Trajectory

Item Type Conference Paper

Author Y. Zhang

Author S. Zhang

Author R. Luo

Abstract Recent years, autonomous driving has always been a hot topic, which is regarded as an important means to shape the future of transport. However, safety is always the essential of any auto-pilot systems in the real-world. The safety and security of autonomous vehicles faces big challenges in public acceptance and trust. Thus, being able to accurately perceive the surrounding environment and respond in real time is an important goal of autonomous driving at the decision-making level. In this paper, a lane change prediction model using convolutional neural network is proposed combining the features from space and time. The proposed CNN learns the behavior pattern from historical data and capable of making lane change predictions to assist the decision making of auto-pilot systems in real-time. The experiments are conducted on the Next Generation Simulation (NGSIM) dataset, and we are able to predict up to 7s horizon before the actual lane-change took place with the observations of one second. An accuracy of 99 % is achieved for a 3 second observation window without further feature engineering. Comparisons are performed with tree based and multi-layer perceptron based methods.

Date 8-12 Oct. 2022

DOI 10.1109/ITSC55140.2022.9921941

Extra Journal Abbreviation: 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)

Pages 287-292

Proceedings Title 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)

Conference Name 2022 IEEE 25th International Conference on Intelligent Transportation Systems (ITSC)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

autonomous vehicles, Safety, Decision making, Lane change, Shape, Real-time systems, Predictive models, Autopilot, convolutional neural network, intention prediction, NGSIM, Time series analysis

Measuring trust in automated driving using a multi-level approach to human factors

Item Type Conference Paper

Author P. Clement

Author H. Danzinger

Author O. Veledar

Author C. Könczöl

Author G. Macher

Author A. Eichberger

Abstract As the driving is shifting towards automation, the maximization of related benefits would profit from improved user acceptance of the new technology. Studies suggest a strong connection between acceptance and trust in technical solutions. We investigate the improvement of user trust to driving automation through demonstrations carried out in a sophisticated driving simulator. The study correlates subjective data with objective psychophysiological measurements. The multi-factorial and multivariate analysis of variance investigates the influence of learning effects and pre-experience with advanced driver assistance systems on trust. Results show improvement in trust through user interaction with a human-machine interface of the demonstrated AD system, hence illustrating the relevance of human-centered development processes. The conclusion is supported by the observation of driver cardiac signals.

Date 1-3 Sept. 2021

DOI 10.1109/DSD53832.2021.00069

Extra Journal Abbreviation: 2021 24th Euromicro Conference on Digital System Design (DSD)

Pages 410-417

Proceedings Title 2021 24th Euromicro Conference on Digital System Design (DSD)

Conference Name 2021 24th Euromicro Conference on Digital System Design (DSD)

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Tags:

Automation, trust, User experience, Machine learning, HMI, Human factors, Particle measurements, Digital systems, ADAS/AD, comfort, Data integrity, driver state, heart-rate, In-cabin measurement, multi-level approach, psychophysiology

MemWAFL: Efficient Model Aggregation for Wireless Ad Hoc Federated Learning in Sparse Dynamic Networks

Item Type Conference Paper

Author K. Eguchi

Author H. Ochiai

Author H. Esaki

Abstract In recent years, privacy-sensitive data has become increasingly prevalent in autonomous vehicles, smart devices, and sensor nodes that can move around and make opportunistic contact with each other. The federation of these nodes has been discussed in the context of federated learning with a centralized mechanism. However, due to multi-vendor issues, relying on a specific server operated by a third party is not desirable in some cases. To address this challenge, wireless ad hoc federated learning (WAFL) has been proposed to realize a fully distributed collaborative machine learning with the nodes physically encountered. WAFL can develop generalized models from non-IID datasets stored in distributed nodes by exchanging and aggregating them with each other over opportunistic node-to-node contacts. Despite these advancements, WAFL's learning speed and accuracy remain relatively slow in sparsely and dynamically connected situations because the model update frequency has to be lower than that in static and frequently connected situations. In this study, we propose MemWAFL, which improves learning accuracy and speed in sparsely and dynamically connected situations. We incorporated a mechanism to store the models of the nodes contacted in the WAFL learning algorithm. As a result, we successfully improved the average learning speed by up to 1209 epochs and the learning accuracy by up to approximately 0.42% compared to the original WAFL in sparsely and dynamically connected situations.

Date 13-15 Nov. 2023

DOI 10.1109/FNWF58287.2023.10520500

Extra Journal Abbreviation: 2023 IEEE Future Networks World Forum (FNWF)

ISBN 2770-7679

Pages 1-5

Proceedings Title 2023 IEEE Future Networks World Forum (FNWF)

Conference Name 2023 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Federated learning, Heuristic algorithms, Collaboration, Data models, Ad hoc networks, Training data, Wireless sensor networks, Ad Hoc networks, Decentralized deep learning, Non-IID data, Peer-to-Peer systems

Mobility IoT -Cars that Think and Communicate-

Item Type Conference Paper

Author Y. Sakagami

Author T. Arakawa**Author** N. Kubota**Author** Y. Toda

Abstract Summary form only. The digital society is rapidly developing due to the IoT that connects all things through the Internet. Information sharing and control will include many issues, create various services, solve issues for people and society, and provide affluence. Information on the activities of various people and information on services are attracting attention as important assets. On the other hand, looking at mobility, the creation of further industries such as neural nets, accelerated development of autonomous driving by improving the performance of computers, the spread of MaaS, the construction of smart cities, services in cities and there, and the use of information in the movement of people. Raw is expected. Therefore, collection and analysis of such information from the viewpoint of mobility, personalization, cooperation with environmental information / traffic information, model of human movement and social activities, concealment and use of personal information, highly safe using information, etc. We would like to hold this workshop as a place to present the latest research results on mobility systems.

Date 26 Oct.-11 Nov. 2022**DOI** 10.1109/WF-IoT54382.2022.10152108**Extra** Journal Abbreviation: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)**Pages** 1-2**Proceedings Title** 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)**Conference Name** 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)**Date Added** 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Autonomous vehicles, Automobiles, Research and development, Human-robot interaction, Internet of Things, Conferences, System analysis and design, personalization, high safety mobility system utilizing the information, information collection and analysis, IoT based mobility systems, modeling of human mobilities and social activities, security and use of personal information

Next Generation Trustworthy Fraud Detection**Item Type** Conference Paper**Author** S. Xie**Author** P. S. Yu

Abstract Popular web applications, such as e-commerce, social networks and online ad auction, are providing valuable services to web users but have also been plagued by prevalent and diverse frauds. Many detection methodologies have been devised but detection trustworthiness is still one important and yet missing desideratum: a user will not trust a detector that has uncertain accuracy, can malfunction under unexpected situations, or can't explain its behaviors and internal working. Previous efforts mostly focused on detection accuracy, and our goal is to chart a path towards a more comprehensive definition of trustworthy detection,

that consists of accuracy, transparency, and proactivity. To achieve the goal, we identify key challenges rooting at the specific settings of the above applications: the evolving nature and unexpectedness of the fraudsters' strategies, the ever-growing large amount of data, and the increasing complexity of effective detectors. We hope spark a large volume of research questions and solutions with respect to the above challenges.

Date 18-20 Oct. 2018

DOI 10.1109/CIC.2018.00045

Extra Journal Abbreviation: 2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC)

Pages 279-282

Proceedings Title 2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC)

Conference Name 2018 IEEE 4th International Conference on Collaboration and Internet Computing (CIC)

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Tags:

Security, Support vector machines, Machine learning, adversarial machine learning, Social network services, Data models, Detectors, fraud detection, Sensitivity analysis

One Bit-Flip Away from Collapse: Security and Safety in AI Models and Accelerators

Item Type Conference Paper

Author S. Das

Author S. Bhattacharya

Author M. H. G. Nejad

Author S. Kundu

Author A. Raha

Author S. Kundu

Author K. Basu

Abstract The unprecedented progress in deep learning has led to the widespread deployment of advanced Artificial Intelligence (AI) models, such as deep neural networks (DNNs), convolutional neural networks (CNNs), transformer-based large language models (LLMs), vision-language models (VLMs), and emerging statespace models like Mamba, across domains including healthcare, autonomous systems, satellite communication, and social media. While these models have achieved remarkable performance, their ever-increasing scale and reliance on specialized Edge AI accelerators (used in real-time, battery-operated systems for ondevice deep learning) have exposed critical safety and security vulnerabilities. In particular, both unintentional hardware faults (e.g., soft errors) and adversarial attacks (e.g., fault injection, bitflips) can lead to severe degradation or complete failure of AI systems, with potentially devastating consequences in real-world edge applications. This paper provides a comprehensive review of bit-flip fault and attack vulnerabilities across the AI stack, spanning model architectures and

hardware platforms. Integrating insights from existing literature with practical assessments, this study intends to be a valuable reference for those dedicated to creating AI systems that are safe, robust, and secure.

Date 18-20 Dec. 2025

DOI 10.1109/ICEdge67252.2025.11412441

Extra Journal Abbreviation: 2025 First International Conference on Intelligent Computing and Systems at the Edge (ICEdge)

Volume 1

Pages 1-7

Proceedings Title 2025 First International Conference on Intelligent Computing and Systems at the Edge (ICEdge)

Conference Name 2025 First International Conference on Intelligent Computing and Systems at the Edge (ICEdge)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Security, Deep learning, Social networking (online), Large language models, Hardware, Transformers, Edge AI, Deep Neural Networks (DNNs), Edge AI Accelerators, In-Memory Computing, Large Language Models (LLMs), Soft Errors, State-space methods, State-Space Models, Systolic Array, Systolic arrays

Optimized Deep Vision Models for High-Dimensional Big Data Interpretation and Visual Understanding

Item Type Conference Paper

Author R. R. Shobitha

Author E. Anbalagan

Author A. A. Ali Samhan

Author A. K. K. Reddy

Author A. R J

Abstract With the rapid expansion of high-dimensional image and video data across various domains—ranging from satellite surveillance and medical imaging to social media and industrial automation—the need for robust, scalable, and interpretable vision models has become critical. Traditional deep learning models, though powerful, often struggle to generalize across such vast data landscapes without significant computational costs and training inefficiencies. This paper proposes an optimized suite of deep vision models specifically designed for interpreting high-dimensional big data and enhancing visual understanding. The proposed framework incorporates advanced convolutional neural network (CNN) architectures, such as EfficientNet, Vision Transformers (ViTs), and hybrid attention-based models, and combines them with big data processing engines like Apache Spark, Kafka, and Hadoop. By adopting intelligent compression techniques, adaptive learning rate scheduling, and distributed training strategies, the models achieve both accuracy and scalability. The system also integrates dimensionality reduction methods such as t-SNE and PCA to mitigate the curse of

dimensionality and improve model convergence. Experiments on large-scale datasets including Open Images, DeepLesion, and Places365 demonstrate the superior interpretability and efficiency of the proposed models. Key performance gains include reduced training times by up to 40%, enhanced classification accuracy by 5–8%, and improved generalization across unseen domains. Applications explored include real-time crowd behavior analysis, automated disease detection, and autonomous navigation. In summary, this work contributes an optimized deep vision modeling framework tailored for big data environments, enabling rapid, accurate, and interpretable visual analytics across high-dimensional data sources.

Date 19-20 Sept. 2025

DOI 10.1109/APCIT65661.2025.11410687

Extra Journal Abbreviation: 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT)

Pages 1-6

Proceedings Title 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT)

Conference Name 2025 2nd Asia Pacific Conference on Innovation in Technology (APCIT)

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Tags:

Technological innovation, Computer vision, Big Data, Computational modeling, Transformers, Accuracy, Training, Adaptation models, Videos, Visual analytics, Big Data Interpretation, Deep Vision Models, Distributed Learning, EfficientNet, High-Dimensional Data, Vision Transformers, Visual Understanding

Optimizing 3GPP 5G NEF Integration: Enhancing Programmable Networks Through NEF and UPF Interoperability

Item Type Conference Paper

Author A. S. L

Author M. Patwary

Abstract This paper investigates a novel approach to enhancing the 3GPP Network Exposure Function (NEF) within 5G core networks, aimed at advancing programmable network capabilities. The research identifies key challenges currently hindering NEF adoption by Communication Service Providers (CSPs), including complexities in user application traffic identification and security concerns associated with CSP-specific Software Development Kits (SDKs). To address these issues, we propose an integration of NEF with the User Plane Function (UPF) at the control plane level. This integration optimizes Business to Customer/Business to Business (B2C/B2B) use cases, such as automatic Quality of Service (QoS) upgrades within a Network as a Service (NaaS) framework. By relying on the direct communication between UPF and NEF—rather than on the 3PP server to provide Traffic Flow Template (TFT) parameters—our approach eliminates the need for 3PP involvement in traffic flow management, significantly reducing delays in QoS adjustments and enhancing real-time network responsiveness. This integration facilitates immediate traffic flow identification at

the network layer, leading to faster QoS provisioning and enhanced resource allocation. The proposed framework demonstrates improvements for 5G and beyond-5G (B5G) deployments, potentially enabling CSPs to optimize service delivery and support network monetization.

Date 15-17 Oct. 2024

DOI 10.1109/FNWF63303.2024.11028837

Extra Journal Abbreviation: 2024 IEEE Future Networks World Forum (FNWF)

ISBN 2770-7679

Pages 423-427

Proceedings Title 2024 IEEE Future Networks World Forum (FNWF)

Conference Name 2024 IEEE Future Networks World Forum (FNWF)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Complexity theory, 5G, Quality of service, Real-time systems, 5G mobile communication, Servers, Business, Interoperability, 3GPP, 5G CORE, 5G Monetization, Beyond5G, IP networks, NaaS API, PCF, QoS, Thin film transistors, UPF

Overseas Gliding UAV Parachute-Free Airdrop Development Status and Trend Research

Item Type Conference Paper

Author J. Guo

Author Y. Guo

Author Z. Wang

Author R. Gao

Abstract The development of glider UAV parachute-less airdrop has brought about a fundamental change in airdrop resupply, realizing the right time, right place, and right amount of resupply of equipment and munitions, and comprehensively enhancing the deployment and guarantee capability of troops at the strategic and tactical levels. In order to promote the research and application of parachute-free airdrop in China, this paper reviews the development status of foreign glider UAV parachute-free airdrop, compares the advantages and limitations of various glider UAVs, and looks forward to the future development trend of glider UAVs, which will be developed in the direction of improving autonomy, intelligence, low cost, stealth, clustering, environmental adaptability, and autonomous takeoff and landing in order to provide a reference for the development of glider UAV parachute-free airdrop technology in China. in order to provide reference for the development of China's glider UAV parachute-less airdrop technology.

Date 26-28 Sept. 2025

DOI 10.1109/LEFICLU65987.2025.11297273

Extra Journal Abbreviation: 2025 Low-Altitude Economy Forum & International Conference on Low-Altitude Flight Technology and Unmanned Aerial Vehicle Application (LEF & ICLU)

Pages 1-6

Proceedings Title 2025 Low-Altitude Economy Forum & International Conference on Low-Altitude Flight Technology and Unmanned Aerial Vehicle Application (LEF & ICLU)
Conference Name 2025 Low-Altitude Economy Forum & International Conference on Low-Altitude Flight Technology and Unmanned Aerial Vehicle Application (LEF & ICLU)
Date Added 20/09/2026, 7:47:04 pm
Modified 20/09/2026, 7:47:04 pm

Tags:

Costs, Reviews, Market research, Weapons, Autonomous aerial vehicles, airdrop resupply, current development status, gliding UAVs, parachute-free airdrops

Parental Trust in Automated Detection of Cyberpredators

Item Type Conference Paper
Author L. Brandimarte
Abstract Online cyberpredators are a serious threat against children who are increasingly using social media and messaging systems to interact with strangers. At the same time, monitoring children's online activity is challenging for parents, due to the numerous platforms and ways a child can access them. Automated approaches could help detect dangerous conversations, but their adoption may prove difficult due to algorithm aversion - the tendency of humans to place less trust in recommendations by automated systems as compared to the judgment of other humans. Three online experiments investigate whether and under what conditions parents are willing to adopt automated systems for detection of cyberpredators, and when they are willing to trade potentially sensitive information about their children's online interaction, as well as individual control and agency. Factors tested for an effect on parental trust include accuracy of predictions by humans versus the algorithm, and storage and management of data in the cloud or local client. Implications for researchers and developers of automated systems of cybercrime detection are presented and discussed.
Date 22-26 May 2023
DOI 10.23919/MIPRO57284.2023.10159713
Extra Journal Abbreviation: 2023 46th MIPRO ICT and Electronics Convention (MIPRO)
ISBN 2623-8764
Pages 30-35
Proceedings Title 2023 46th MIPRO ICT and Electronics Convention (MIPRO)
Conference Name 2023 46th MIPRO ICT and Electronics Convention (MIPRO)
Date Added 20/09/2026, 7:47:04 pm
Modified 20/09/2026, 7:47:04 pm

Tags:

Safety, Social networking (online), Computer crime, Prediction algorithms, Oral communication, Information and communication technology, algorithmic aversion, Communication channels, cyberpredators, experiments

Predictive Control of Diesel Oxidation Catalysts with Federated Learning in Connected Vehicles

Item Type Conference Paper

Author V. P. Chellapandi

Author Y. Nagaraj

Author J. Supplee

Author S. Hernandez-Gonzalez

Author H. Borhan

Author S. H. Žak

Abstract In various industries, diesel engines continue to be indispensable due to their exceptional fuel efficiency and torque characteristics, all while meeting stringent emissions standards through advanced aftertreatment technologies like Diesel Oxidation Catalysts (DOC), Diesel Particulate Filters (DPF), and Selective Catalytic Reduction (SCR). The critical role of hydrocarbon dosing in Diesel Oxidation Catalysts (DOC) lies in effectively reducing emissions such as hydrocarbons, carbon monoxide, and soluble organic fractions in particulate matter, converting them into H₂O and CO₂. However, traditional DOC control strategies often struggle to adapt to the dynamic nature of diesel engine operation, resulting in temperature overshoots during transient engine conditions. To address this challenge, an approach is proposed that utilizes Long Short-Term Memory (LSTM) networks within a Federated Learning (FL) framework to predict and control DOC temperature overshoots in realtime, enabling adjustments to the hydrocarbon dosing strategy. This approach not only aims to minimize overshoots but also extends the lifespan of aftertreatment components and reduces emissions. Moreover, when applied to connected vehicles, Federated Learning ensures rigorous data privacy by conducting model training locally within each vehicle, preserving the confidentiality of sensitive information while fostering collaborative improvements in emissions control.

Date 26-28 Feb. 2024

DOI 10.1109/FISTS60717.2024.10485594

Extra Journal Abbreviation: 2024 Forum for Innovative Sustainable Transportation Systems (FISTS)

Pages 1-6

Proceedings Title 2024 Forum for Innovative Sustainable Transportation Systems (FISTS)

Conference Name 2024 Forum for Innovative Sustainable Transportation Systems (FISTS)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Federated learning, Data models, Training, Catalysts, Hydrocarbons, Oxidation, Temperature distribution

Privacy Preserving and Federated Learning-based Drowsy Driving Detection for ITS

Item Type Conference Paper

Author A. Sakhrelia

Author A. A. Padaria

Author N. K. Jadav

Author R. Gupta

Author S. Tanwar

Author N. Yamsani

Abstract The issue of drowsy driving is highly dangerous for road safety and lead to car accidents, injuries as well as deaths. This study examines the several technology suggestions made to address the serious issue of drowsy driving, which compromises road safety worldwide. The study looks at different sleep-related accident prevention tactics and assesses how well they work with intelligent transportation systems (ITS). Federated Learning (FL) is an intriguing method that ensures privacy while improving accuracy for identifying weariness. Furthermore, the paper offers a novel method of transparently responsible traffic control by fusing blockchain technology with convolutional neural networks (CNNs). This integrated system authenticates smart contracts, maintains stringent records, and quickly detects drowsiness to protect system integrity. Essentially, these results highlight effective learning dynamics in FL, like strong aggregation techniques. Furthermore, this fully integrated architecture is not only exceptionally good at detecting sleep, but it also boasts a safe and dependable environment that is applicable to a variety of sectors in terms of international settings and the dependability of academic discourse.

Date 29-31 Aug. 2024

DOI 10.1109/ICEECT61758.2024.10739331

Extra Journal Abbreviation: 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)

Volume 1

Pages 1-6

Proceedings Title 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)

Conference Name 2024 International Conference on Electrical Electronics and Computing Technologies (ICEECT)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Vehicles, Road safety, Privacy, Traffic control, Intelligent transportation systems, Convolutional neural networks, Federated learning, deep learning, Smart contracts, federated learning, Computers, and intelligent transportation systems, blockchain technology, convolutional neural networks, driver drowsiness detection, Injuries

R-LiViT: A LiDAR-Visual-Thermal Dataset Enabling Vulnerable Road User Focused Roadside Perception

Item Type Conference Paper

Author J. Mirlach
Author L. Wan
Author A. Wiedholz
Author H. E. Keen
Author A. Eich

Abstract In autonomous driving, the integration of roadside perception systems is essential for overcoming occlusion challenges and enhancing the safety of Vulnerable Road Users (VRUs). While LiDAR and visual (RGB) sensors are commonly used, thermal imaging remains underrepresented in datasets, despite its acknowledged advantages for VRU detection in extreme lighting conditions. In this paper, we present R-LiViT, the first dataset to combine LiDAR, RGB, and thermal imaging from a roadside perspective, with a strong focus on VRUs. R-LiViT captures three intersections during both day and night, ensuring a diverse dataset. It includes 10,000 LiDAR frames and 2,400 temporally and spatially aligned RGB and thermal images across 150 traffic scenarios, with 7 and 8 annotated classes respectively, providing a comprehensive resource for tasks such as object detection and tracking. The dataset11<https://doi-org.ezproxy.lib.uts.edu.au/10.5281/zenodo.16356714> and the code for reproducing our evaluation results22<https://github.com/XITASO/r-livit> are made publicly available.

Date 19-25 Oct. 2025

DOI 10.1109/ICCV51701.2025.02635

Extra Journal Abbreviation: 2025 IEEE/CVF International Conference on Computer Vision (ICCV)

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Pages 28375-28384

Proceedings Title 2025 IEEE/CVF International Conference on Computer Vision (ICCV)

Conference Name 2025 IEEE/CVF International Conference on Computer Vision (ICCV)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

autonomous driving, Vehicle-to-everything, Communication systems, Protocols, Sensor systems, camera, HTTP, 3d perception, Aerospace and electronic systems, Computer networks, dataset, lidar, Location awareness, Mobile communication, roadside perception, thermal camera, Wide area networks

RB-XAI: Relevance-Based Explainable AI for Traffic Detection in Autonomous Systems

Item Type Conference Paper

Author I. Adom

Author M. N. Mahmoud

Abstract Recent progress in artificial intelligence has brought about a revolutionary era spanning various sectors such as transportation, healthcare, finance, and cybersecurity. In the field of transportation, these advancements empower autonomous systems to precisely perceive their surroundings, make swift and

secure decisions, and operate seamlessly without human intervention. In the case of Autonomous Vehicles (AVs), the perception phase, involving tasks like road surface extraction and on-road object detection, is crucial. While AVs offer benefits like improved road safety, convenience, and efficiency, they grapple with the “black box” challenge, symbolizing the opacity of their decision-making processes to humans. This absence of transparency poses acceptance, regulatory, ethical, and security issues. To tackle these challenges, our work leverages Concept Relevance Propagation (CRP), a bias-resistant Relevance-Based (RB) eXplainable Artificial Intelligence (XAI) algorithm to proffer transparent concept - level explanations to the behavior of object detection models used in AVs, for traffic perception. CRP goes beyond traditional attribution maps, by generating explanations that automatically identifies and visualizes relevant concepts within the input space. This insight sheds light on the crucial areas responsible for the behavior of object detection models used in AVs. This research aims to boost transparency, trust, and understanding in the field of AVs, fostering safer and more widely accepted autonomous systems.

Date 15-24 March 2024

DOI 10.1109/SoutheastCon52093.2024.10500215

Extra Journal Abbreviation: SoutheastCon 2024

ISBN 1558-058X

Pages 1358-1367

Proceedings Title SoutheastCon 2024

Conference Name SoutheastCon 2024

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Autonomous Vehicles, Road safety, Transportation, Regulation, Visualization, Object detection, Concept Relevance Propagation, eXplainable Artificial Intelligence, Object Detection, Explainable AI, Robustness

Real-Time Analytics for Big Data in Smart City Applications: Transforming Urban Environments with Data-Driven Insights

Item Type Conference Paper

Author A. Khanderi

Author G. Rajaram

Author R. R. Hussein

Author P. Johri

Author T. Mohanraj

Author M. Balamurugan

Abstract This paper offers an overview of real-time analytics for smart cities proposing a framework for real-time data processing to improve the urban landscape. In the proposed system, large-scale data-streams can be processed using advanced machine learning algorithms, edge computing, and distributed data storage. Analyzing the outcomes of the evaluation procedure evidences reportable

performance enhancements in the areas of smart city applications. Average traffic speed in transportation systems in the same period rose by 60%, concurrency and response to incidents likewise cut down by 40% and 53% respectively. The energy management systems responded with a 16% decrease in maximum energy load and the waste reduction was by 50%, while the costs savings was by 25% increase. Public safety applications alone realised a 60% decrease in the time taken to detect a crime and a 28.6% increase in efficiency on disaster resources with an overall spike in public satisfaction of 25.7%. Such findings improve the understanding of how this framework can increase urban efficiency, sustainability, and quality of life, which makes the solution usable in future smart city projects.

Date 4-6 March 2025

DOI 10.1109/AUTOCOM64127.2025.10956868

Extra Journal Abbreviation: 2025 International Conference on Automation and Computation (AUTOCOM)

Pages 302-306

Proceedings Title 2025 International Conference on Automation and Computation (AUTOCOM)

Conference Name 2025 International Conference on Automation and Computation (AUTOCOM)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Costs, Smart cities, Big Data, Vehicle dynamics, Real-time systems, Smart City, Congestion Reduction, Data-Driven Insights, Disasters, Energy consumption, Energy Demand Reduction, Energy management systems, Memory, Performance Metrics, Real-Time Analytics, Sustainable Urban Development, Traffic Speed, Urban Environment, Waste reduction

Recent Trends on Artificial Intelligence-Enabled Internet of Things Platform and Standard Technologies

Item Type Conference Paper

Author J. Lee

Author A. Abid

Author F. Le Gall

Author J. Song

Abstract Artificial intelligence (AI), Internet of Things (IoT), and cloud computing are regarded as the core technologies of a variety of state-of-the-art technologies having disruptive innovation that are intended to lead the 4th Industrial Revolution. A common point shared by the services that utilise the three technologies, such as smart city, autonomous vehicles, unmanned aerial vehicles, and smart factory, is that they are all services related to data. That is, data lie at the heart of future technologies. However, there are still challenges in the development of efficient services and technologies due to the absence of standards for data collected in a variety of domains. Therefore, in this study, we examine the latest trends in the development of international standards for the management of data in support of AI services and technologies led by oneM2M,

which is a global initiative for the development of international standards for IoT and cloud platforms. More specifically, the trends include the development of common functions in support of AI technology, standard technologies for the support of data licence management, support for data compliance with privacy protection regulations, and semantic functions for metadata management in support of AI/machine learning.

Date 26 Oct.-11 Nov. 2022

DOI 10.1109/WF-IoT54382.2022.10152226

Extra Journal Abbreviation: 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

Pages 1-6

Proceedings Title 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

Conference Name 2022 IEEE 8th World Forum on Internet of Things (WF-IoT)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

AI, machine learning, Regulation, Smart cities, Semantics, Internet of Things, Cloud computing, Market research, Training data, data licence management, IoT platform, oneM2M

Recognition of Facial Expressions and Emotions on the Face using the YOLOv5 Algorithm

Item Type Conference Paper

Author A. A. S. Gunawan

Author M. R. Kusnadi

Author J. T. Joseph

Author W. Josan

Author I. S. Edbert

Abstract This study examines the significant influence of emotional states on individuals, recognizing that people are often confronted with numerous challenges and problems in their lives. Such challenges can give rise to feelings of despair and tension, especially in those who struggle to manage them effectively. Depression, which is not just limited to adults, can also have a significant impact on children who may not fully grasp their socio-economic conditions. One of the serious consequences of unresolved emotional distress, particularly depression, is an elevated risk of suicide. Within this framework, facial recognition technology presents itself as a powerful instrument for identifying human emotions. The purpose of this study is to assess the efficacy of facial recognition systems in detecting indicators of depression. Using the YOLOv5 algorithm and comparing it with its precursors, YOLOv4 and YOLOv7, the research conducts a simulation experiment to examine the ability of the system to precisely interpret a range of emotional states, including happiness, sadness, and anger. The research findings suggest that the YOLO algorithm, particularly in its most recent versions, displays exceptional precision in identifying and classifying intricate facial expressions, thus showcasing its potential as a beneficial device in mental health evaluations.

Date 1-2 Dec. 2023**DOI** 10.1109/IWAIIP58158.2023.10462848**Extra** Journal Abbreviation: 2023 International Workshop on Artificial Intelligence and Image Processing (IWAIP)**Pages** 337-341**Proceedings Title** 2023 International Workshop on Artificial Intelligence and Image Processing (IWAIP)**Conference Name** 2023 International Workshop on Artificial Intelligence and Image Processing (IWAIP)**Date Added** 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Facial Recognition, Emotion recognition, Face recognition, Heuristic algorithms, Facial Expressions, Emotion, Face Detection, Training, YOLO, Streaming media, Depression, Mental Disorders, Stressful

Research on Automated Guided Vehicles Path Planning Based on Improved A-star

Item Type Conference Paper**Author** L. Zhang**Author** K. Xue**Author** L. Zhang**Author** X. Zhang**Author** H. Yang

Abstract To address the issues of path redundancy, frequent turning, and low task allocation efficiency in multi-AGV path planning for automotive warehousing environments, this paper proposes a cooperative path planning method that integrates the Grey Wolf Optimizer (GWO) with a penalty-function-based A-star algorithm. First, a dynamic penalty function mechanism is introduced into the traditional A-star algorithm, where penalty weights are applied to AGV steering actions to effectively suppress unnecessary turns, thus improving path smoothness and operational efficiency. A bidirectional search strategy is also adopted to reduce computational complexity. Second, the multi-target task allocation problem is modeled as a Traveling Salesman Problem (TSP) and globally optimized using an enhanced Grey Wolf Optimizer. By incorporating a nonlinear convergence factor and a dynamic weight strategy, the algorithm's ability to escape local optima is improved, allowing for rapid generation of the optimal task sequence. Experiments conducted on a warehouse map show that, compared with the traditional A-star algorithm, the proposed method offers significant advantages in path length, number of unnecessary turns, and overall task completion time. This method greatly enhances the transportation reliability, safety, and operational efficiency of AGV systems in automotive warehousing, providing effective technical support for the intelligent operation of high-density warehousing logistics.

Date 7-9 Nov. 2025**DOI** 10.1109/IFEEA66847.2025.11388591

Extra Journal Abbreviation: 2025 12th International Forum on Electrical Engineering and Automation (IFEEA)
Pages 349-356
Proceedings Title 2025 12th International Forum on Electrical Engineering and Automation (IFEEA)
Conference Name 2025 12th International Forum on Electrical Engineering and Automation (IFEEA)
Date Added 20/09/2026, 7:47:04 pm
Modified 20/09/2026, 7:47:04 pm

Tags:

Transportation, Heuristic algorithms, Vehicle dynamics, Resource management, A-star Algorithm, AGV Path Planning, Improved Grey Wolf Optimizer, Path planning, Penalty Weight, Redundancy, Search problems, Traveling salesman problems, Turning, Warehousing

Resilient Scheduling of Real-Time Cyber-Physical Systems Against Memory-Corruptions

Item Type Conference Paper

Author A. Al Arafat

Author K. Wilson

Author S. Vaidhun

Author B. C. Ward

Author Z. Guo

Abstract Real-time cyber-physical systems (CPS) are increasingly deployed in command and control applications for safety and mission-critical domains such as autonomous vehicles and critical infrastructure. To enable enhanced capabilities, CPS are becoming more complex and interconnected, yet this expanded functionality introduces new security vulnerabilities. Addressing these challenges, this paper presents a secure and resilient scheduling technique for hard real-time CPS applications that protects against common memory-corruption-based attacks. Our approach introduces a security-oriented dimension of criticality, enabling the system to selectively drop low-security-critical workloads in response to detected threats. This reduces the attack surface and allows for the timely rescheduling of both victim task re-executions and system recovery processes. We demonstrate that traditional mixed-criticality scheduling approaches are overly conservative and inadequate for accommodating dynamic recovery requirements under this security model. To address this, we propose a novel scheduling algorithm tailored for security-aware CPS, along with a schedulability test using a security-criticality demand-bound function. The proposed framework is implemented in FreeRTOS with micro-ROS and validated using a hardware-in-the-loop simulation of a flight control task. Extensive schedulability experiments reveal that our model outperforms existing approaches with required adaptation, improving acceptance ratios by over 30 percent in heavily utilized CPS environments. This work advances secure, real-time scheduling to enhance both the resilience and safety of critical cyber-physical applications.

Date 20-22 Aug. 2025
DOI 10.1109/RTCSA66114.2025.00023
Extra Journal Abbreviation: 2025 IEEE 31st International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)
ISBN 2325-1301
Pages 138-149
Proceedings Title 2025 IEEE 31st International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)
Conference Name 2025 IEEE 31st International Conference on Embedded and Real-Time Computing Systems and Applications (RTCSA)
Date Added 20/09/2026, 7:47:03 pm
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Tags:

Safety, Security, Cyber-physical systems, mixedcriticality systems, real-time cyber-physical system, resilient scheduling, secure recovery, Vehicle dynamics, Real-time systems, Surface treatment, Adaptation models, Dynamic scheduling, Scheduling algorithms, System recovery

Secure CAN for Connected Vehicles

Item Type Conference Paper
Author T. Dee
Author A. Tyagi
Abstract A Controller Area Network (CAN) bus facilitates real time data/command transfer over small distances. It is frequently used in automotive, off-road, and industrial environments with several IoT nodes connected to it. Control messages with commands can cause an actuator to perform some action. An adversary might observe the control message and subsequent action. Injecting the control message at a later time (replay attack) should not replay the same action. Unfortunately current CAN protocol provides no protection against such attacks. Third party components may reverse engineer messaging protocols to gain control over critical systems resulting in dangerous system malfunctions. In a connected vehicle ecosystem, these attacks can also originate at peer vehicles with transactions coming in through one of the IoT nodes on the CAN bus. The proposed solution is to verify the integrity and authenticity of messages. Verifying the integrity of a freshness value appended to messages prevents replay attacks. Authenticity guarantees messages do not come from a malicious party. A keyed hash function using a shared secret key enables authenticated messages. The challenges are: (1) establishing a shared secret and (2) synchronizing a freshness value. Past standards limit CAN message sizes to 8 bytes. New standards (CAN-FD) allow for longer messages. We propose appending 8 bytes to messages to provide integrity and authenticity. Nodes require a shared key and freshness value for creating these bytes. Many scenarios require shared key renegotiation. The shared key is negotiated using Elliptic Curve Cryptography (ECC) infrastructure. Protocols are developed for establishing a shared key and synchronizing a freshness value. Protocol correctness is verified through simulations.
Date 2-16 June 2020

DOI 10.1109/WF-IoT48130.2020.9221400
Extra Journal Abbreviation: 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Pages 1-6
Proceedings Title 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Conference Name 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Date Added 20/09/2026, 7:47:03 pm
Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Security, Connected vehicles, Real-time systems, Protocols, Peer-to-peer computing, Elliptic curve cryptography

Semantic Information Market For The Metaverse: An Auction Based Approach

Item Type Conference Paper

Author L. Ismail

Author D. Niyato

Author S. Sun

Author D. I. Kim

Author M. Erol-Kantarci

Author C. Miao

Abstract In this paper, we address the networking and communications problems of creating a digital copy in the Metaverse digital twin. Specifically, a virtual service provider (VSP) which is responsible for creating and rendering the Metaverse, is required to use the data collected by IoT devices to create the virtual copy of the physical world. However, due to the huge volume of the collected data by IoT devices (e.g., images and videos) and the limited bandwidth, the VSP might become unable to retrieve all the required data from the physical world. Furthermore, the Metaverse needs fast replication (e.g., rendering) of the digital copy adding more restrictions on the data transmission delay. To solve the aforementioned challenges, we propose to equip the IoT devices with semantic information extraction algorithms to minimize the size of the transmitted data over the wireless channels. Since many IoT devices will be interested to sell their semantic information to the VSP, we propose a truthful reverse auction mechanism that helps the VSP select only IoT devices that can improve the quality of its virtual copy of objects through the semantic information. We conduct extensive simulations on a dataset that contains synchronized camera and radar images, and show that our novel design enables a fast replication of the digital copy with high accuracy.

Date 10-14 Oct. 2022

DOI 10.1109/FNWF55208.2022.00115

Extra Journal Abbreviation: 2022 IEEE Future Networks World Forum (FNWF)

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Pages 628-633

Proceedings Title 2022 IEEE Future Networks World Forum (FNWF)

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Tags:

Semantics, Wireless networks, Information retrieval, Training, Rendering (computer graphics), Digital twins, Metaverse, reverse auction, semantic communication, sensing-as-a-service

Smart Self-Diagnosis Method for GPS Attacks and Safety Faults in UAVs

Item Type Conference Paper

Author I. G. Ferrão

Author A. L. De Oliveira

Author D. Espes

Author C. Dezan

Author K. C. Branco

Abstract Unmanned aerial vehicles (UAVs) have been used in a variety of applications in safety-critical domains such as logistics, transportation, and defense. However, safety assurance is needed for the widespread development and operation of UAVs in urban areas where a failure may lead to physical harm to people, environment, or property. Since UAV s are highly interconnected, security is an important concern for the acceptance of safety-critical UAV applications. Thus, it is needed to ensure that UAV s are protected from attacks, e.g., Global Position System (GPS) signal spoofing, and unauthorized manipulation of flight controls, caused by malicious external agents aiming to compromise system/data confidentiality, integrity, or availability. Artificial Intelligence techniques such as Machine Learning classifiers can be used to support the identification of UAV s attacks. This study presents the development of safety and security diagnostic methods for UAVs. Two multiclass approaches are proposed, focusing on identifying safety-related failures and security threats such as GPS spoofing and jamming attacks. Various machine learning algorithms are employed, and evaluation metrics demonstrate high accuracy rates across models. The study underscores the importance of addressing both safety and security aspects in UAV s operations and provides pre-processing code for reproducibility. The developed methods contribute to enhancing UAV resilience and understanding safety and security measures in unmanned aerial operations.

Date 4-7 June 2024

DOI 10.1109/ICUAS60882.2024.10556910

Extra Journal Abbreviation: 2024 International Conference on Unmanned Aircraft Systems (ICUAS)

ISBN 2575-7296

Pages 1301-1308

Proceedings Title 2024 International Conference on Unmanned Aircraft Systems (ICUAS)

Conference Name 2024 International Conference on Unmanned Aircraft Systems (ICUAS)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Transportation, Machine learning, Machine learning algorithms, Urban areas, Autonomous aerial vehicles, Reproducibility of results

Standard Latent Space Dimension for Network Intrusion Detection Systems Datasets

Item Type Journal Article

Author R. F. Moyano

Author A. Duque

Author D. Riofrío

Author N. Pérez

Author D. Benitez

Author M. Baldeon-Calisto

Author D. Fernández

Abstract Machine learning is a branch of artificial intelligence that provides computers the ability to create or improve algorithms without being explicitly programmed by directly learning from data. It is widely used in automation or decision-making tasks in fields such as image or speech recognition, sentiment analysis, or self-driving cars. However, its application in the field of communication networks is limited by the lack of appropriate research resources, such as rich datasets for training or the definition of a standard set of features. In this context, a standard latent space dimension is proposed by performing an autoencoder-based dimensionality reduction process. Different network security datasets are projected onto a lower-dimensional space to determine a standard or convergent dimension. The convergent dimension is determined by identifying the threshold above which diminishing returns begin to occur in the autoencoder loss as the latent space dimension increases. The experimental validation showed that four machine learning classification models, trained with a standard latent space of ten dimensions, performed as well as the models that used the non-reduced versions of the datasets in terms of F1-score and accuracy. Furthermore, a Wilcoxon statistical test showed that the mean accuracy of all classification models trained with the standard latent space dimension had a difference of less than 0.0235 in comparison to the models trained with the original inputs. A negligible difference in accuracy is a significant outcome because researchers can use only the latent space to perform experiments with certainty that the performance of ML models will not be constrained.

Date 2023

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Journal Abbr IEEE Access

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Tags:

Artificial intelligence, Feature extraction, autoencoder, Dimensionality reduction, latent space, machine learning, Network intrusion detection, network security, Standardization, Machine learning, Telecommunication traffic, Communication networks

Technical Factors Matter for Driving Behavior Analysis: Experiment and Validation

Item Type Journal Article

Author Y. Wei

Author W. Wang

Author C. Zhang

Author J. Xi

Abstract Technological factors like driver assistance systems have the potential to influence human driving behavior, forming their driving habits. However, the precise mechanisms underlying the impact of these technological factors on behavior remain elusive. This paper specifically targets drive modes (i.e., eco and sport) as the technological factors and explores their influence on humans' driving behavior. First, we collect naturalistic and simulation driving data from 16 drivers who operated vehicles in eco and sport modes. We then employ eight hypothesis tests to analyze the changes in driving behavior with eco and sport modes. This study reveals that human drivers' behavior preferences are related to the level of urgency of the driving task that needs to be completed, with observable behavior characteristics being contingent upon both the drive mode and the complexity of traffic situations. We experimentally speculate that single-drive modes such as eco or sport mode cannot adapt to the full range of driving needs due to differences in the level of urgency of driving tasks to be completed, which is further validated with practical applications in simulation. These findings provide the basis for improving the acceptance of human-centric driver assistance systems for different vehicles.

Date July 2025

DOI 10.1109/TIV.2024.3471255

Volume 10

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Publication IEEE Transactions on Intelligent Vehicles

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Journal Abbr IEEE Transactions on Intelligent Vehicles

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Tags:

Autonomous vehicles, Security, Simulation, Data collection, driving behavior, Driver behavior, Vehicle dynamics, Testing, Behavior preference, Damping, hypothesis testing, technical factors

Telecommunication & Networking ♦ Changing Customer Profile & Preferences

Item Type Conference Paper

Author A. Mehrotra

Author S. Menon

Abstract Over the last two and half decades, the Indian telecom industry has evolved rapidly. Beginning with the basic telephony provided by BSNL, a public sector company, the telecom industry today is not just for providing the calling facilities; it includes services like 3G, 4G, 5G broadband, Netflix, amazon-prime, online payment banks, 24 hour recharge facilities voicemail, ringtones, easy bill payment services and much more. Internet is just one of the important additions to the services. While many big players in the private sector came on the scene and disappeared, others struggled to survive through mergers, acquisitions and restructuring of the business models in the wake of competition. Sometime back Reliance Jio offered free Internet and Voice usage as pre ♦ launch offer and grabbed the customer's attention. This all came as a big game changer in the telecom industry as customers have multiple options and the telephonic and data service providers face new challenges to cope up with the situation. It is estimated that a large number of new customers coming into the mobile fold come from the underprivileged geographies with sub-optimal telecommunication infrastructure and poor connectivity; their prime preference is for low call tariffs and several other value added services like information about markets, commodity prices, weather forecasts and the ability to transact business in their regional languages. The theme of the paper is to examine if the telecommunication industry is prepared to face and respond to changing customer perceptions, preferences and expectations.

Date 19-21 Jan. 2021

DOI 10.1109/ICCAKM50778.2021.9357766

Extra Journal Abbreviation: 2021 2nd International Conference on Computation, Automation and Knowledge Management (ICCAKM)

Pages 221-226

Proceedings Title 2021 2nd International Conference on Computation, Automation and Knowledge Management (ICCAKM)

Conference Name 2021 2nd International Conference on Computation, Automation and Knowledge Management (ICCAKM)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

5G, Weather forecasting, Industries, Telecommunications, Faces, Internet, Business, Communications technology, 4G, Airtel, BSNL, Customer Preference, Reliance Jio Effect, Telecommunication

The emergence of ethics engineering in Industrial Cyber-Physical Systems

Item Type Conference Paper

Author P. Leitão

Author S. Karnouskos

Abstract Artificial Intelligence (AI) is being successfully adopted in innovative industrial applications in different sectors, e.g., manufacturing, logistics, healthcare, smart grids, and smart transportation. However, the potential direct or indirect threats of using AI are not sufficiently investigated in these domains, and as a result, unwanted vulnerabilities are introduced in industrial applications, e.g., related to business info leakage, privacy, and even undesired/false training and learning of personnel and automated processes. The ethical dimension is new ground for most industrial applications and poses a significant challenge in AI systems overall, and especially for complex Industrial Cyber-Physical Systems (ICPS), which are based on collective intelligence and exhibit autonomy, emergence and self-organization. This work discusses the importance and implications of ethics in industrial AI-based systems such as the ICPS. For this purpose, the concept of ethics in such intelligent systems is surveyed and characterized using a bibliometric literature review, and then, areas linked to ethics, as well as the main challenges for engineering ethically compliant ICPS are discussed.

Date 24-26 May 2022

DOI 10.1109/ICPS51978.2022.9816931

Extra Journal Abbreviation: 2022 IEEE 5th International Conference on Industrial Cyber-Physical Systems (ICPS)

Pages 1-6

Proceedings Title 2022 IEEE 5th International Conference on Industrial Cyber-Physical Systems (ICPS)

Conference Name 2022 IEEE 5th International Conference on Industrial Cyber-Physical Systems (ICPS)

Date Added 20/09/2026, 7:47:03 pm

Modified 20/09/2026, 7:47:03 pm

The Impact of AI on Human Psychology: A User Perspective

Item Type Conference Paper

Author A. Mohan

Author D. Singh

Author A. Kataria

Author H. Kumar Meena

Author R. Singh

Abstract The rapid integration of artificial intelligence into many human activities has produced a growing conversation about its psychological consequences on users. The main focus of the comprehensive responses of AI users in this research is the cognitive and emotive aspects of its influence on human psychology. Using the data collected from a varied group of users, this study exposes the influence of AI interactions on user perceptions, emotions, and mental processes. Key results show that although artificial intelligence technologies can significantly boost productivity, inspire innovation, and simplify decision-making, they can also cause anxiety, reliance, and changed social behaviors. The paper investigates essential concerns, including the psychological consequences of deepfake technology, the possibility of artificial intelligence recording human rights violations, and the consequences of AI-driven monitoring. It stresses both

positive and negative effects. Furthermore, under evaluation are consumers' degrees of adaptation to the growing influence of artificial intelligence and the variations in psychological reactions among different user groups. This study underlines the need for transparency, user autonomy, and ethical AI deployment by pointing out ways that AI developers and stakeholders might apply to lower unexpected psychological effects. This thorough investigation helps to clarify how artificial intelligence shapes human psychological experiences and supports a sensible approach that maximizes benefits and reduces possible risks. The research aims to guide the path of following artificial intelligence developments, thereby ensuring technologically conscious and user-centric technological innovations.

Date 16-18 Jan. 2025

DOI 10.1109/IC3ECSBHI63591.2025.10990736

Extra Journal Abbreviation: 2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)

Pages 1321-1326

Proceedings Title 2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)

Conference Name 2025 International Conference on Cognitive Computing in Engineering, Communications, Sciences and Biomedical Health Informatics (IC3ECSBHI)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

Safety, Data privacy, Artificial intelligence, AI, Reliability, Technological innovation, Surveys, Psychology, Shape, Stakeholders, AI User, Behavior, Feelings, Human Psychology, Recording, Thinking, Unintended Impacts of AI

The Impact of Artificial Intelligence on the Labor Market in the Arab Gulf Region

Item Type Conference Paper

Author A. J. AlMahdawi

Author E. A. Zaitoun

Author M. S. Jaradat

Author M. E. Elzeiny

Abstract This study aimed to investigate the impact of artificial intelligence (AI) on the labor market by exploring workers' perspectives on AI's positive and negative impacts and society's trends in coping with these impacts. Using a survey-based questionnaire, data were collected from 162 individuals, including both employed and unemployed people with various educational qualifications. The results indicated that AI has both positive and negative impacts to a high extent, with mean scores of 3.85 for positive impacts and 3.75 for negative impacts. Significant differences were observed based on gender, educational qualifications, and professional roles: males (Mean $\bar{X}$ =4.2, SD =0.9) perceived AI's impact more positively than females (Mean = 3.6, SD = 1.1), while higher

educational qualifications were associated with a more optimistic view of AI (e.g., Master's Degree holders had a Mean $\$=4.4$, $\mathrm{SD}=0.8$). Additionally, freelancers (Mean $\$=4.0$, $\mathrm{SD}=1.1$) viewed AI as both a threat and opportunity, whereas employees (Mean $\$=3.7$, $\mathrm{SD}=1.0$) and job seekers (Mean $\$=3.3$, $\mathrm{SD}=1.2$) had varying concerns regarding job security and skill demands. Overall, the high acceptance of AI reflects a positive trend towards technology in the labor market.

Date 24-27 Sept. 2024

DOI 10.1109/IDSTA62194.2024.10746940

Extra Journal Abbreviation: 2024 Fifth International Conference on Intelligent Data Science Technologies and Applications (IDSTA)

Pages 197-202

Proceedings Title 2024 Fifth International Conference on Intelligent Data Science Technologies and Applications (IDSTA)

Conference Name 2024 Fifth International Conference on Intelligent Data Science Technologies and Applications (IDSTA)

Date Added 20/09/2026, 7:47:04 pm

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Tags:

Security, Artificial intelligence, Artificial Intelligence, impact, Market research, Data science, labor market, Oman, Qualifications, UAE

The Optimized Algorithm Creation for the Whole HE System

Item Type Conference Paper

Author M. Nazirov

Author N. Qolqanov

Abstract Thus, integration of artificial intelligence (AI) has become one of the very critical forces in the molding organizational dynamics within the present rapidly changing modern business environment. This paper, therefore, discourses a very large area: leadership in the age of AI, whose dense interconnections exist between technological development and organizational strategy and capacity in human leadership. Thus, the development of AI, according to the study, brings a different aspect of operations from decision-making processes right down to interfacing with customers, and leaders are to grapple with the challenges and opportunities that this presents as it demands strategic adaptation and innovative approaches. This paper sets out to explain how the technological paradigm AI is shifting the conventional paradigms of leadership. The very nature of their work is literally changed from being just managers to orchestrating collaboration between human and machine because of how AI-driven automation makes routine tasks either really easy and self-explanatory or because of the analytical capabilities of the tasks super critical to that transformation is developing a leadership ethos that recognizes the value and stewardship of adaptive standards for continuous learning and ethical use of AI technologies. This paper, therefore, discusses the ethical dilemmas that are evidently present in AI adoption in organizational contexts. These range from data protection and algorithmic bias to ethical implications in AI-based decision-making, cumulatively summing up to a

rather tortuous ethical jungle through which the leaders should guide their organizations to a responsible deployment of AI technologies.

Date 14-15 May 2024

DOI 10.1109/ICACITE60783.2024.10616409

Extra Journal Abbreviation: 2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)

Pages 1588-1592

Proceedings Title 2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)

Conference Name 2024 4th International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE)

Date Added 20/09/2026, 7:47:04 pm

Modified 20/09/2026, 7:47:04 pm

Tags:

algorithmic bias, Decision making, data privacy, Technological innovation, Artificial intelligence (AI), Reviews, decision-making, Leadership, Organizations, Standards organizations, Ethics, ethical considerations, organizational success, skills gap, transformational leadership

The Rise of AI Communication: Opportunities, Challenges, and the Path Forward

Item Type Conference Paper

Author P. Lin

Abstract Artificial intelligence (AI) has emerged as a transformative technology, displaying enormous promise and impact across many areas. Communication, a pillar of modern-day society, has also begun adopting applications of AI. This paper investigates the advancement of AI-powered communication by examining the technology, current uses, and future outlook. It sheds light on AI communication's deep implications for the communication sphere and broader society. On the technology front, specialized AI techniques for communication tasks have been rolled out, including machine translation, neural networks and more. Application-Wise, AI communication already underpins intelligent assistants, information filtering and personalized recommendations. Looking ahead, AI communications could drive more customized, interactive and smart communication experiences, while also raising pressing privacy and ethical considerations needing urgent attention. Overall, AI-charged communication is profoundly penetrating and reshaping communication markets and people's lifestyles.

Date 11-13 June 2025

DOI 10.1109/BMSB65076.2025.11165632

Extra Journal Abbreviation: 2025 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)

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Pages 1-5

Proceedings Title 2025 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)
Conference Name 2025 IEEE International Symposium on Broadband Multimedia Systems and Broadcasting (BMSB)
Date Added 20/09/2026, 7:47:03 pm
Modified 20/09/2026, 7:47:03 pm

Tags:

Artificial intelligence, Privacy, Communication, Neural networks, Pressing, Ethics, Information filtering, Adopting AI applications, Advancement of AI-powered communication, Broadband communication, Broadcasting, Machine translation, Multimedia communication, Promise and impact, Transformative technology

The Use of Machine Learning in the Automotive Industry

Item Type Conference Paper
Author Y. Shrivastava
Author A. K. Sethi
Author O. Vikash
Abstract Machine Learning (ML) is a subfield of artificial intelligence, which is already making a difference to the automotive industry by contributing towards overall performance, safety and user experience. Applications of machine learning in various fields include production, autonomous riding, and predictive remodeling — this evaluation. In autonomous driving, particularly, machine learning algorithms—deeper learning algorithms—to enhance vision, decision making and management capabilities equipping cars to safely negotiate challenging terrain. Machine leaning (ML) is used to use predictive preservation to reduce downtime, lower costs, and increase protection by predicting when a car may breakdown. An example of where Machine Learning aids in good manufacturing house with great control and reduces production process and therefore increases productivity and cuts down costs. In addition, machine learning (ML) through natural language processing (NLP), provides for more personalized services and better in car aides which increases the consumer satisfaction. Despite its many positive impacts, ML poses challenges for the automobile industry, which is: privacy concerns; legal compliance; and technology. The conclusion of this essay is that while these circ ones present significant challenges to meet, the future of machine learning in the automotive industry looks great — and who knows what enormous impacts machine learning might have on efficiency and personalization.
Date 20-21 Sept. 2024
DOI 10.1109/I3CEET61722.2024.10994036
Extra Journal Abbreviation: 2024 International Conference on Communication, Computing and Energy Efficient Technologies (I3CEET)
Pages 904-910
Proceedings Title 2024 International Conference on Communication, Computing and Energy Efficient Technologies (I3CEET)
Conference Name 2024 International Conference on Communication, Computing and Energy Efficient Technologies (I3CEET)

Date Added 20/09/2026, 7:47:03 pm**Modified** 20/09/2026, 7:47:03 pm**Tags:**

Data privacy, Automobiles, Machine learning, Costs, Machine learning algorithms, Automotive engineering, Prediction algorithms, Predictive maintenance, Data Privacy, Autonomous Driving, Industries, Process control, Customer Experience (CX), Machine Learning (ML), Predictive Maintenance

Toward Keyword Extraction in Constrained Information Retrieval in Vehicle Social Network

Item Type Journal Article**Author** Z. Yang**Author** H. Yu**Author** J. Tang**Author** H. Liu

Abstract With the emergence of various vehicular apps for automotive navigation, driving safety, and in-car entertainment system, vehicular social data is starting to explode. Such an explosion has posed a great challenge for a centralized pattern of computing, storing, and managing information, calling for a more decentralized (cloud-based) style of processing data for vehicular social network. Decentralized information processing requires encryption of sensitive information prior to out-sourcing for the purpose of protecting data from unsolicited access and thus a searchable keyword index needs to be extracted from the encrypted document for solicited access. However, most existing searchable encryption schemes, based on the words in the subject or content, suffers from poor retrieval performance. In this study, a novel keyword extraction metric based on spatial distribution of a particular text is proposed with the view to improving the retrieval performance. In the proposed solution, constrained information retrieval in vehicle social network is conceptualized as retrieval of data with extremely strict security and privacy preserving policies and a cloud-based vehicle social information retrieval framework is developed. In line with the framework, experiments have been conducted on benchmark data sets and diverse evaluation metrics. The results suggest that the solution proposed in this paper is capable of a wider application and a better performance.

Date May 2019**DOI** 10.1109/TVT.2019.2906799**Volume** 68**Pages** 4285-4294**Publication** IEEE Transactions on Vehicular Technology**ISSN** 1939-9359**Issue** 5**Journal Abbr** IEEE Transactions on Vehicular Technology**Date Added** 20/09/2026, 7:47:04 pm**Modified** 20/09/2026, 7:47:04 pm

Tags:

Social networking (online), Cloud computing, Encryption, Indexes, Constrained information retrieval, content analysis, keyword extraction, vehicle social network

Towards Safety in Automotive Ethernet-based Networks with Dynamic Workloads

Item Type	Conference Paper
Author	A. Kostrzewa
Author	D. Stöhrmann
Author	R. Ernst
Abstract	The advent of automated and autonomous driving requires decision-making based on high-resolution sensor data. In addition, these systems must react safely and promptly to dynamically changing environmental conditions, e.g. traffic situation, weather etc. In this paper we present the frequently considered isolation mechanisms, standards and concepts for achieving functional safety in automotive systems e.g. 802.1Q, AVB, TSN. However, static configurations, which are introduced by the majority of them, penalizes dynamic communication profiles. Consequently, static network management is no longer sufficient to achieve the required performance level. We show that standardised does not necessarily mean safe "out of the box" nor performant and that the new design vectors must be taken into consideration.
Date	2-16 June 2020
DOI	10.1109/WF-IoT48130.2020.9221313
Extra	Journal Abbreviation: 2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Pages	1-6
Proceedings Title	2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Conference Name	2020 IEEE 6th World Forum on Internet of Things (WF-IoT)
Date Added	20/09/2026, 7:47:04 pm
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Tags:

Autonomous vehicles, Safety, Decision making, Internet of Things, Vehicle dynamics, Standards, Meteorology

Trust-oriented HMI Design for Conditional Autonomous Driving Takeover Systems

Item Type	Conference Paper
Author	Q. Sun
Author	S. Hou
Author	J. Yao

Author P. Cheng

Abstract Conditional Autonomous Driving (CAD) will become widespread in the next few years, but the driving features of human-vehicle co-driving could mean significant safety and trust issues. A good cooperation between human and vehicles requires the driver to trust the vehicle appropriately, otherwise it will lead to misuse or disuse of the vehicle by the driver. In this study, based on the theoretical model of trust in automation through literature review, we conducted semi-structured interviews with eight drivers (male=4, female=4) to collect drivers' information needs in the full phase of CAD and preference of existing HMI design. The data was organized and coded based on the interview results. We sorted out the priority of the first 6 HMI messages in pre-takeover, in-takeover and after-takeover stages respectively. The findings were applied to the visual (dashboard and center control screen) and audio HMI in typical takeover scenarios to increase driver's trust in the autonomous driving takeover system and improve driver acceptance and security perception of CAD vehicles.

Date 21-23 Oct. 2022

DOI 10.1109/ICID57362.2022.9969690

Extra Journal Abbreviation: 2022 3rd International Conference on Intelligent Design (ICID)

Pages 196-199

Proceedings Title 2022 3rd International Conference on Intelligent Design (ICID)

Conference Name 2022 3rd International Conference on Intelligent Design (ICID)

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Modified 20/09/2026, 7:47:03 pm

Tags:

Safety, Security, Automation, Visualization, trust in automation, Psychology, Bibliographies, Solid modeling, conditional autonomous driving, HMI design, takeover system

Vehicle Directional Driving Behavior Segmentation Based on Cornering Strength

Item Type Journal Article

Author Y. Bo

Author X. Jia

Author H. Guan

Author H. Wang

Abstract With the rapid development in recent years, the autonomous driving system comes to the stage of commercial deployment now. However, it is not well accepted by society. This is due to the fact that the driving behavior features of autonomous driving vehicles are currently different from those of human drivers. It is important to understand how people drive for increasing acceptance. And the primary task of that is driving behavior identification. In this study, we proposed a segmentation method for vehicle driving process into basic directional driving behaviors. In order to clarify the relationship between vehicle motion state and driver operation, we rebuilt the vehicle motion state in the natural coordinate system. We innovatively proposed the concept of cornering strength as a metric of directional driving behavior intensity. To

verify the validity of our approach, 22 participants were asked to maneuver a car through some common driving scenarios including turn, lane change, overtaking and U-turn. As a result, our method can extract directional driving behaviors with an average accuracy of 93.33%. In addition, the cornering strength of participants with different driving styles exhibit a significant difference though in the same driving scenario. Driving age has a significant effect on CS stability of participants. The variance of CS reaches a maximum of 0.22. The directional driving behavior features of the same participant can be different influenced by psychological or physiological aspects. Road adhesion condition and traffic congestion condition also have an effect on the CS values of participants.

Date 2024

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Journal Abbr IEEE Access

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Tags:

Safety, Vehicles, Autonomous driving, Data collection, Target tracking, Anthropomorphism, Autonomous driving system, Behavioral sciences, Long short term memory, Trajectory, cornering strength, driving behavior segmentation, maneuvering target tracking, Motion segmentation, motion state rebuilding

Video Face Anonymization for Preserving Privacy

Item Type Conference Paper

Author G. Megala

Author R. Venkatesan

Author T. Vigneshwaran

Abstract This work scales with the development of preserving privacy on face anonymization in a video. The aim of this work is to identify, preserve and protect the privacy of individuals who appear in videos with sensitive contexts or wish to conceal their identities before sharing on social media platforms to avoid cyber bullying, cyber stalking and other problems caused by the trend of viral videos in today's social climate. An automatic face detection is implemented and blurring for processing the video. The experimental results shows that person face privacy is preserved temporal in appearance across video frames.

Date 8-10 Dec. 2022

DOI 10.1109/ICDSAAI55433.2022.10028937

Extra Journal Abbreviation: 2022 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)

Volume 01

Pages 1-5

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Conference Name 2022 International Conference on Data Science, Agents & Artificial Intelligence (ICDSAAI)
Date Added 20/09/2026, 7:47:03 pm
Modified 20/09/2026, 7:47:03 pm

Tags:

Data privacy, Deep learning, Privacy, Feature extraction, Visualization, Face recognition, Anonymization, Convolutional Neural Network, ImageNet, Spatial databases

Wireless Technologies for the Next-Generation Train Control and Monitoring System

Item Type Conference Paper

Author J. Härri

Author A. Arriola

Author P. Aljama

Author I. Lopez

Author U. Fuhr

Author M. Straub

Abstract The Next Generation Train Control and Monitoring System (NG-TCMS) represents a major innovation of the European Railway industry introducing a wireless architecture as an enabler to enhanced automation, flexible train management and increased railway capacity. This paper surveys various available and future wireless technologies matching the requirements both at train backbone and consist levels. Illustrating the challenges for a single technology to match all requirements, this paper also highlights the benefits of the upcoming 5G technology for future NG-TCMS.

Date 30 Sept.-2 Oct. 2019

DOI 10.1109/5GWF.2019.8911648

Extra Journal Abbreviation: 2019 IEEE 2nd 5G World Forum (5GWF)

Pages 179-184

Proceedings Title 2019 IEEE 2nd 5G World Forum (5GWF)

Conference Name 2019 IEEE 2nd 5G World Forum (5GWF)

Date Added 20/09/2026, 7:47:03 pm

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Tags:

5G-V2X, 5G, Vehicle-to-everything, Wireless communication, 5G mobile communication, Communication system security, Ethernet, Long Term Evolution, LTE-V2X, NG-TCMS, Rail transportation, Wireless Consist Network, Wireless Train Backbone

Zone-of-Interest Strategy for the Creation of High-Resolution Adversarial Images Against Convolutional Neural Networks

Item Type Conference Paper

Author F. Leprévost

Author A. O. Topal

Author E. Mancellari

Author K. Lavangnananda

Abstract Trained convolutional neural networks (CNNs) are among the leading tools used for the automatic classification of images. They are nevertheless exposed to attacks: Given an input clean image classified by a CNN in a category, carefully designed adversarial images may lead CNNs to erroneous classifications, although humans would still classify “correctly” the constructed adversarial images in the same category as the input image. Currently most attacks are performed in the image input size domain of the considered CNN, which is usually small. However, due to privacy concerns with personal images on social media, there is a demand for generating large adversarial images that preserve the visual information of original images with the highest possible quality, while preventing automatic tracking and personal identification. Creating large-size adversarial images is difficult due to speed, adversity, and visual quality challenges, in particular if a requirement on adversarial images is the inability for humans to notice any difference between them and the original clean images. This paper describes the zone-of-interest generic strategy that aims at increasing drastically the efficiency of any type of attack (white-box or black-box, untargeted or targeted) and any specific attack (FGSM, PGD, BIM, SimBA, AdvGAN, EA-based attacks, etc.) on CNNs. Instead of exploring the full image size, the strategy identifies zones on which to focus the attacks. Although applying to any image size, the strategy is especially valuable for large high-resolution images. This strategy can be combined with other generic approaches, like the noise blowing-up method, to further improve attacks' performances.

Date 26-27 Oct. 2023

DOI 10.1109/ICITEE59582.2023.10317668

Extra Journal Abbreviation: 2023 15th International Conference on Information Technology and Electrical Engineering (ICITEE)

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Tags:

Privacy, Visualization, Social networking (online), Electrical engineering, Image recognition, Convolutional Neural Network, Black-box attack, Closed box, High resolution adversarial image, Image resolution, Noise Blowing-Up method